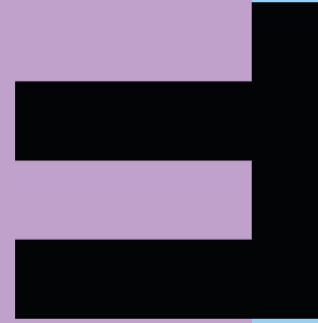


FHV

Vorarlberg University
of Applied Sciences



074703043202 - Energy transmission and distribution

Ruben Lliuyacc Blas

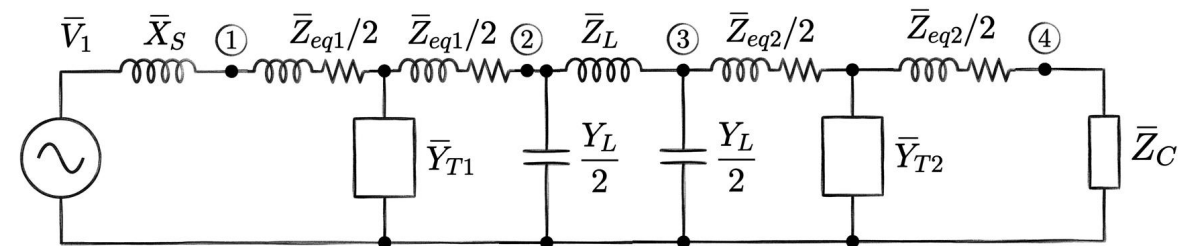
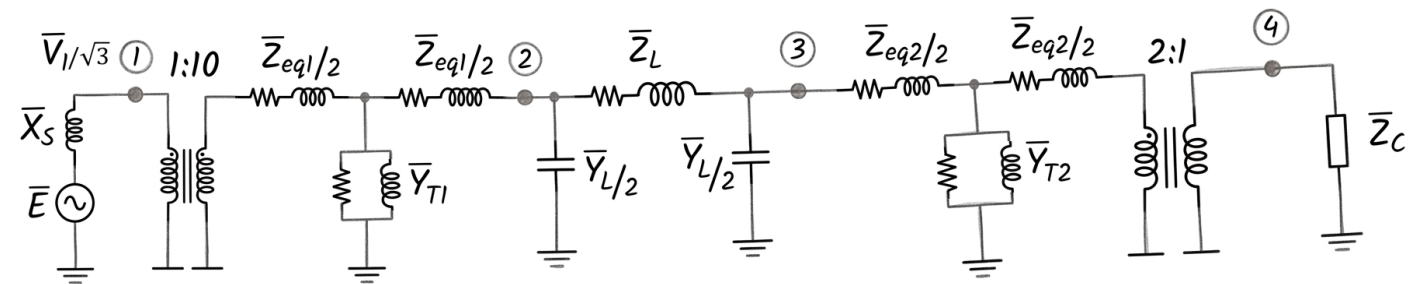
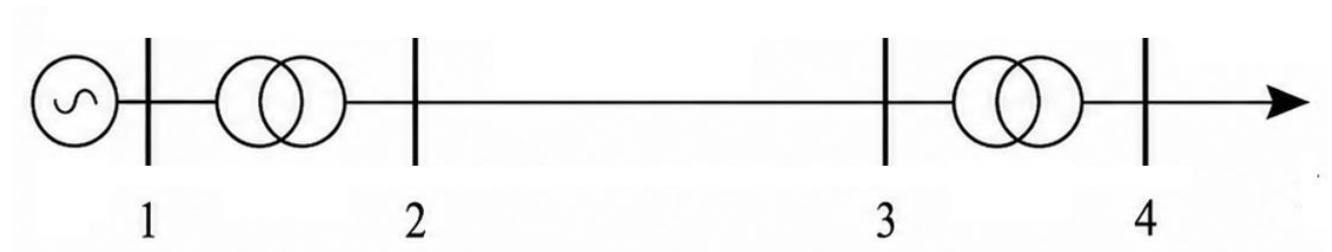
01.06.2026

Recap

❑ **Models of a**
generator, transformer,
and transmission line.

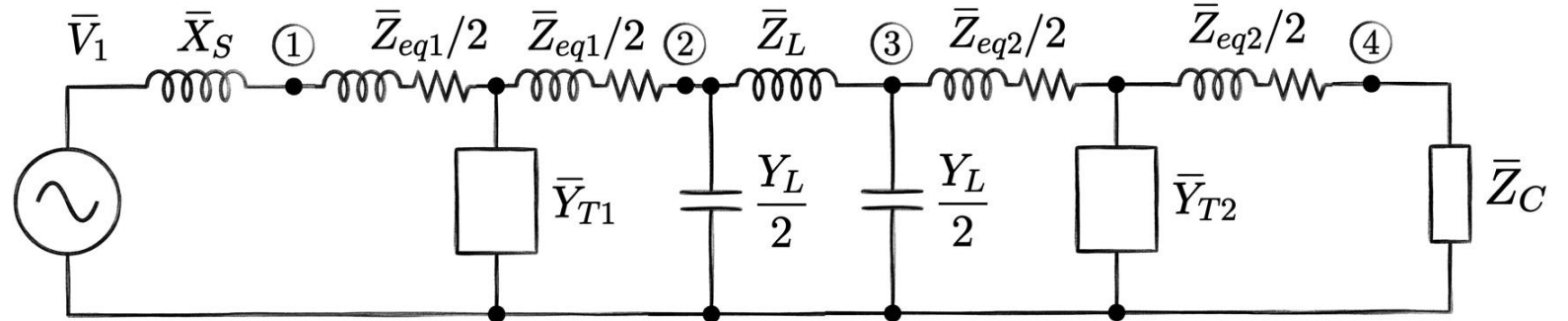
❑ **Single-phase**
equivalent circuit.

❑ **Per-unit** equivalent
circuit.

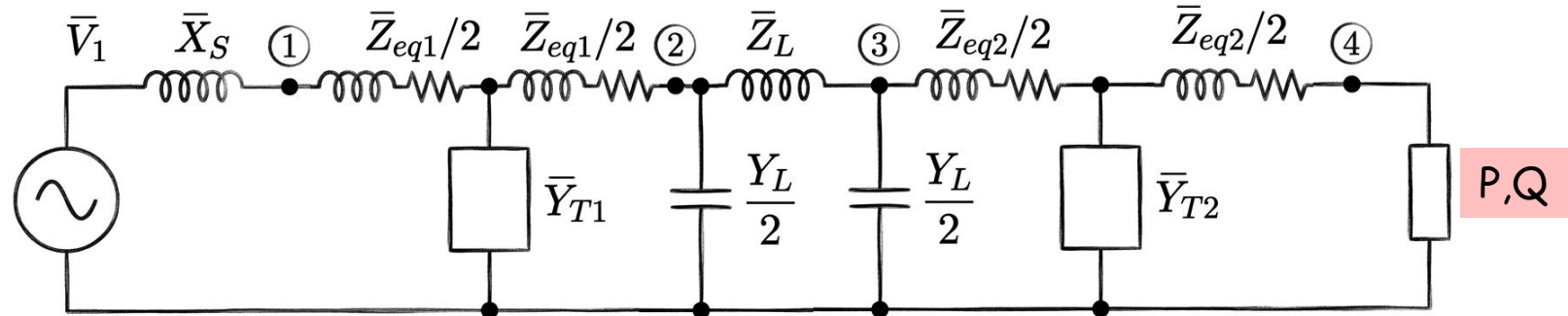


Power flow motivation

☐ Linear

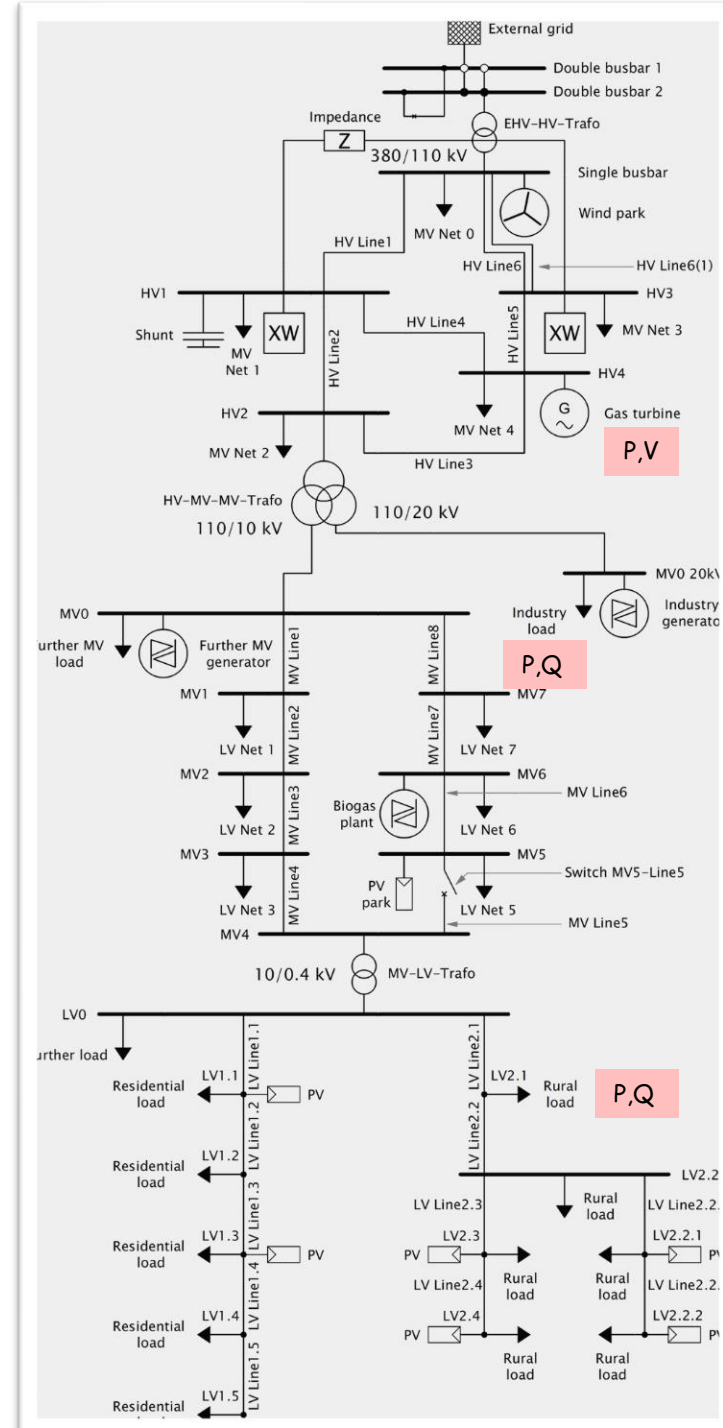


☐ Non-linear



Power flow motivation

- ❑ Active and reactive power instead of impedances.
 - ❑ Active power and voltage module instead of „unlimited power“.
- Large set of **non-linear equations**.
- **Iterative methods**.



Part II: Power flow

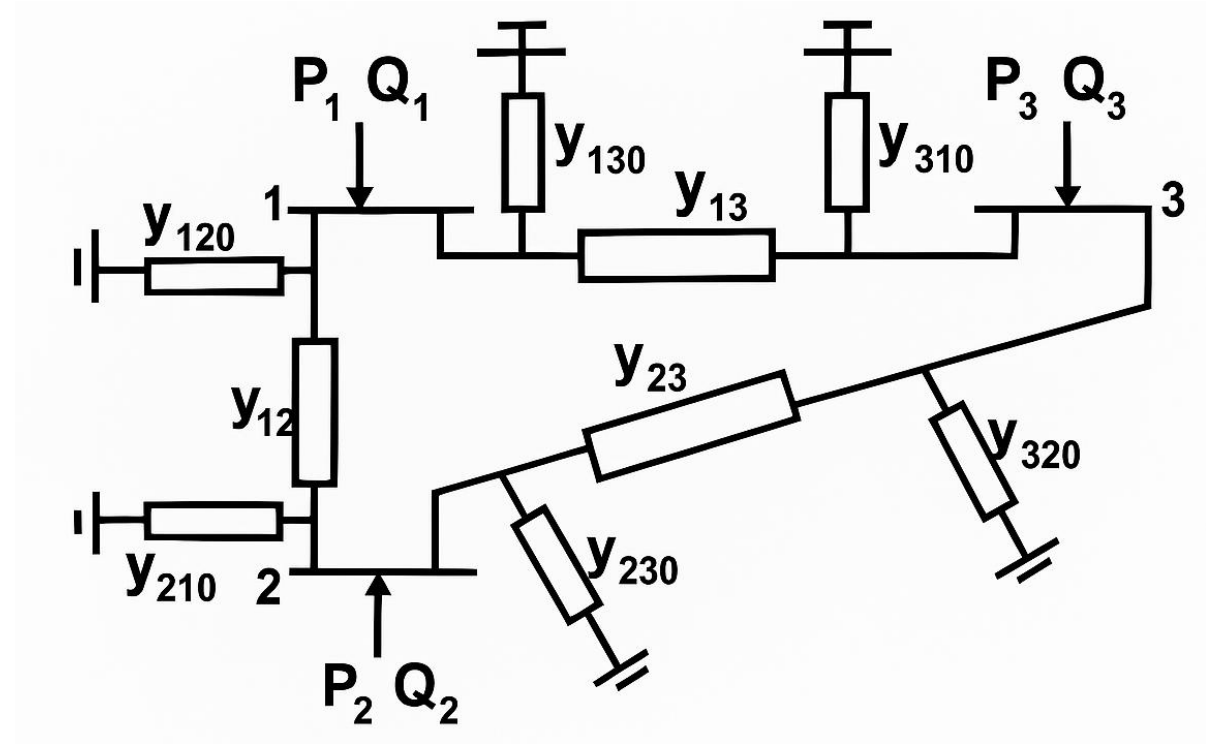
- Admittance Matrix
- Jacobian Matrix
- Solve non-linear equations (power flow equations)

→ **Steady state of the network**

3-bus power system

This system represents:

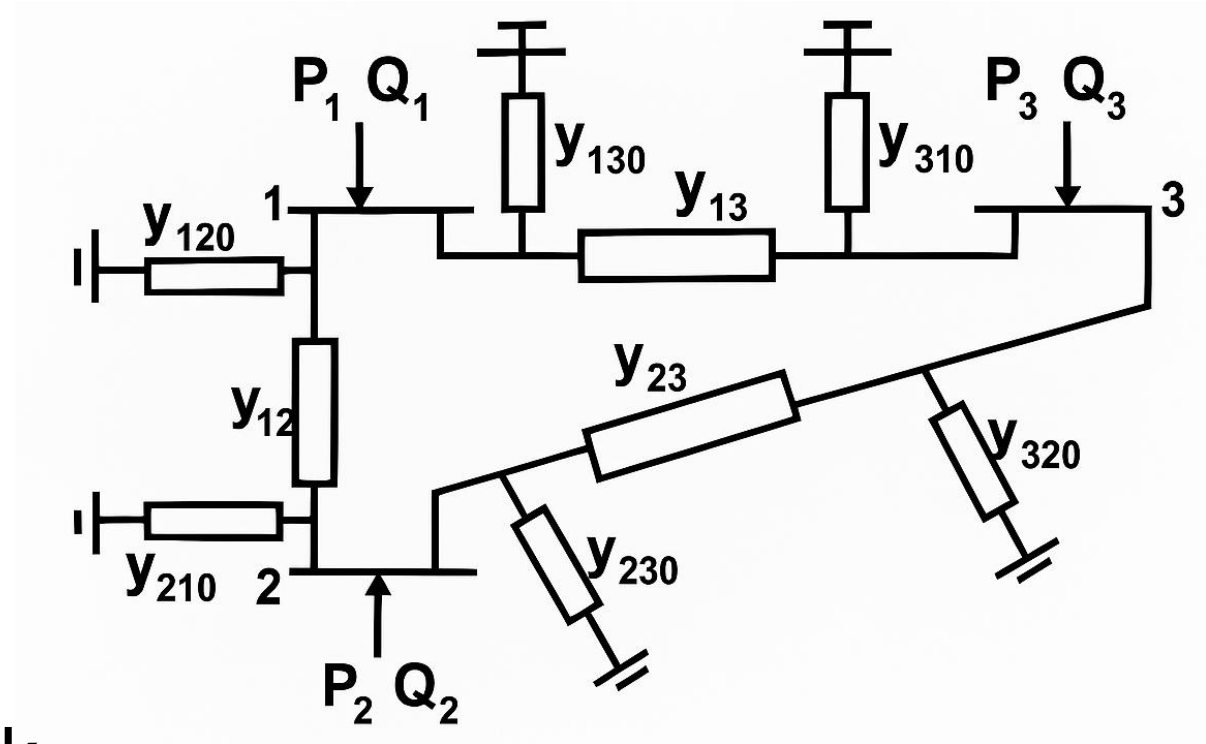
- ❑ A three-phase **balanced** network.
- ❑ Transmission lines are modelled as Π -model.
- ❑ Power injection are positive values (sign convention).



Power flow

Aim:

- to know **all voltages**
(**magnitude** and **angle**)
→ power flow through lines,
transformers.
→ steady-state of the network



Admittance matrix (Y)

Kirchhoff's current law

$$\bar{i}_1 = \bar{y}_{120} \bar{v}_1 + \bar{y}_{12} (\bar{v}_1 - \bar{v}_2) + \bar{y}_{130} \bar{v}_1 + \bar{y}_{13} (\bar{v}_1 - \bar{v}_3),$$

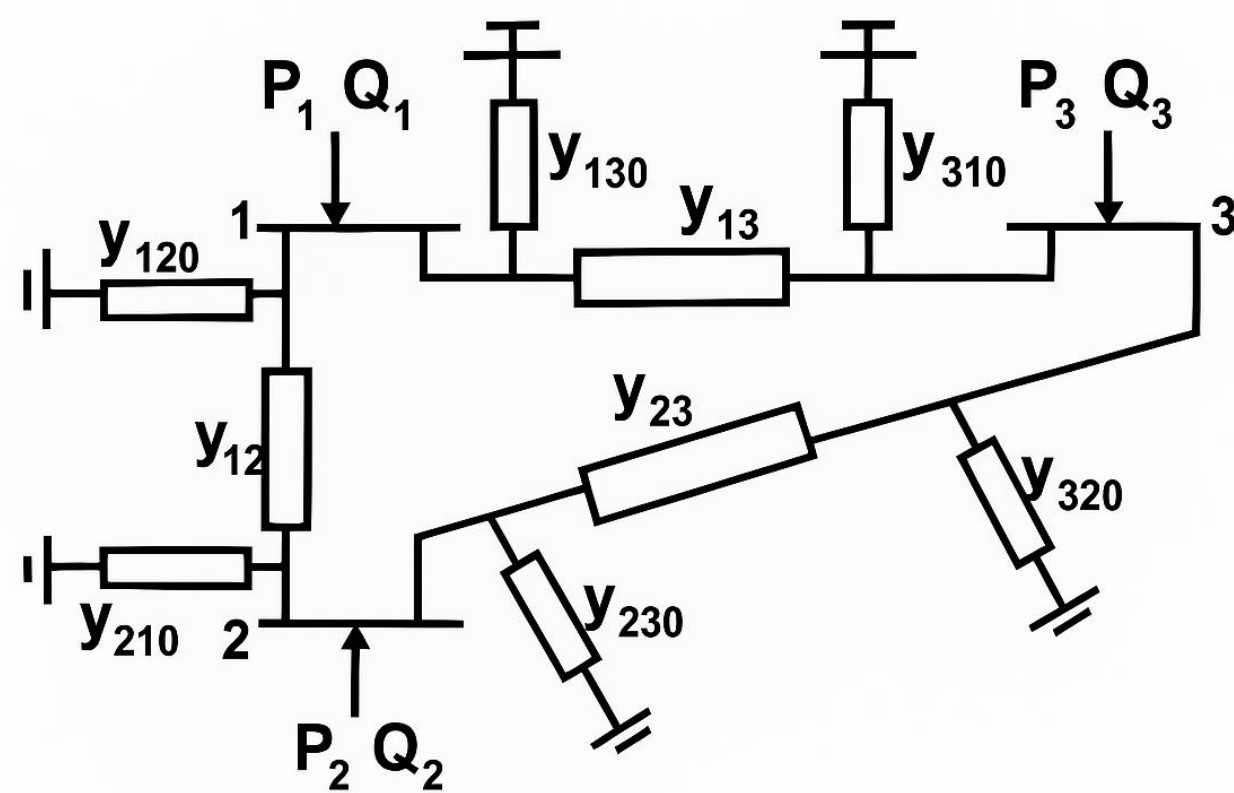
$$\bar{i}_2 = \bar{y}_{210} \bar{v}_2 + \bar{y}_{12} (\bar{v}_2 - \bar{v}_1) + \bar{y}_{230} \bar{v}_2 + \bar{y}_{23} (\bar{v}_2 - \bar{v}_3),$$

$$\bar{i}_3 = \bar{y}_{310} \bar{v}_3 + \bar{y}_{13} (\bar{v}_3 - \bar{v}_1) + \bar{y}_{320} \bar{v}_3 + \bar{y}_{23} (\bar{v}_3 - \bar{v}_2).$$

$$\begin{bmatrix} \bar{i}_1 \\ \bar{i}_2 \\ \bar{i}_3 \end{bmatrix} = \begin{bmatrix} \bar{y}_{120} + \bar{y}_{130} + \bar{y}_{12} + \bar{y}_{13} & -\bar{y}_{12} & -\bar{y}_{13} \\ -\bar{y}_{12} & \bar{y}_{210} + \bar{y}_{230} + \bar{y}_{12} + \bar{y}_{23} & -\bar{y}_{23} \\ -\bar{y}_{13} & -\bar{y}_{23} & \bar{y}_{310} + \bar{y}_{320} + \bar{y}_{13} + \bar{y}_{23} \end{bmatrix} \begin{bmatrix} \bar{v}_1 \\ \bar{v}_2 \\ \bar{v}_3 \end{bmatrix}$$

$$\begin{bmatrix} \bar{i}_1 \\ \bar{i}_2 \\ \bar{i}_3 \end{bmatrix} = \begin{bmatrix} \bar{Y}_{11} & \bar{Y}_{12} & \bar{Y}_{13} \\ \bar{Y}_{21} & \bar{Y}_{22} & \bar{Y}_{23} \\ \bar{Y}_{31} & \bar{Y}_{32} & \bar{Y}_{33} \end{bmatrix} \begin{bmatrix} \bar{v}_1 \\ \bar{v}_2 \\ \bar{v}_3 \end{bmatrix}$$

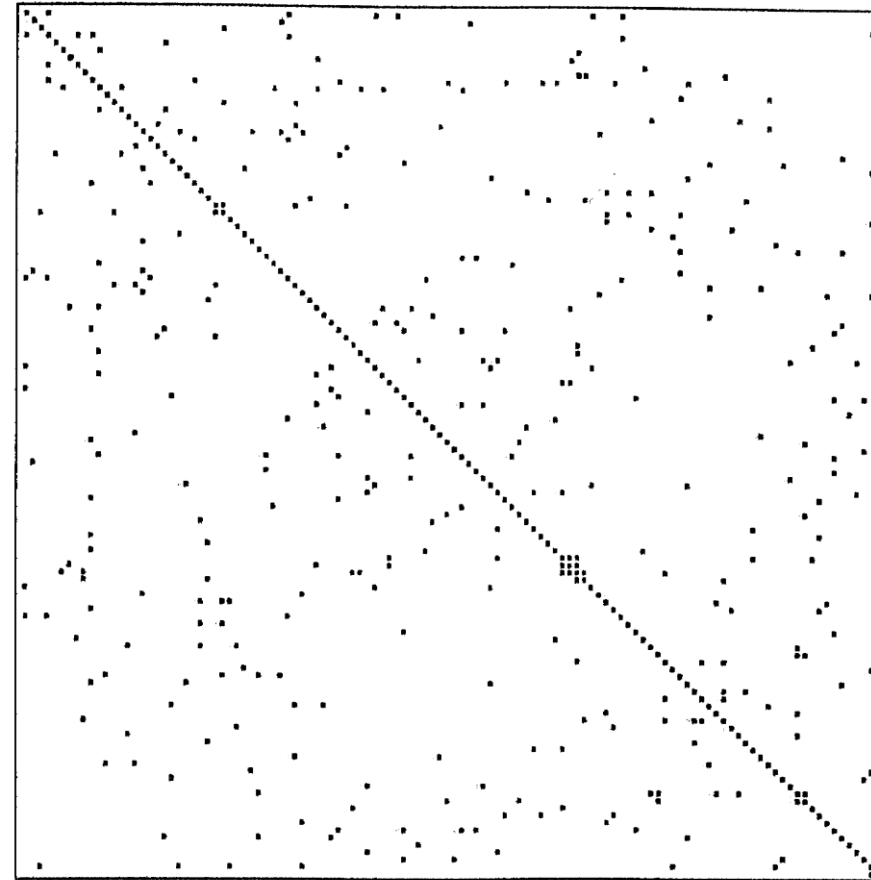
$$\bar{I} = \bar{Y} \bar{V}$$



Y: Admittance matrix

Admittance matrix (Y)

- ❑ $N \times N$, where N = number of buses.
- ❑ Y is a symmetric matrix.
- ❑ Y is a sparse matrix.



(118 rows, 179x2 off-diagonal elements)



Power flow equations

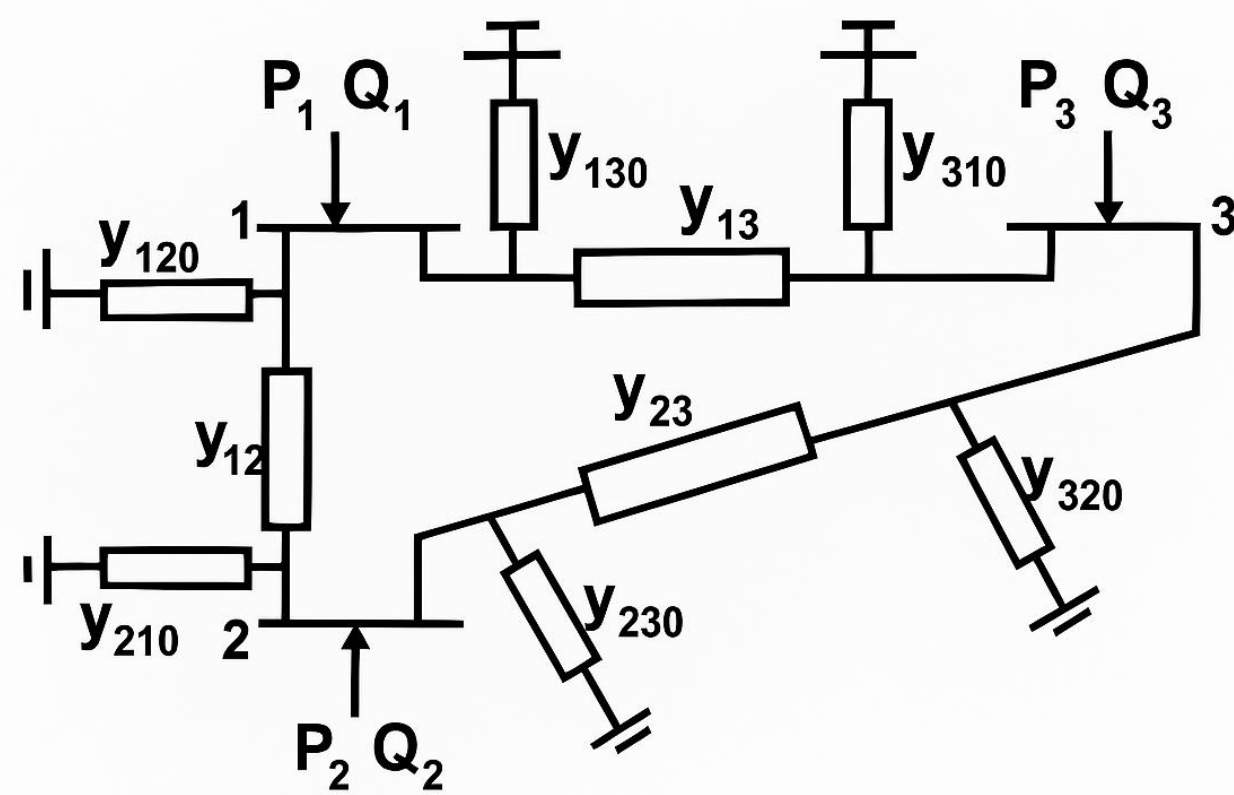
$$\bar{s}_i = p_i + j q_i = \bar{v}_i \bar{i}_i^*$$

$$\bar{i}_i = \sum_k \bar{Y}_{ik} \bar{v}_k$$

$$\bar{s}_i = \bar{v}_i \left(\sum_{k=1}^3 \bar{Y}_{ik} \bar{v}_k \right)^* = \bar{v}_i \sum_{k=1}^3 \bar{Y}_{ik}^* \bar{v}_k^*$$

$$\bar{s}_i = \sum_{k=1}^3 v_i v_k (G_{ik} - j B_{ik}) e^{j(\theta_i - \theta_k)}$$

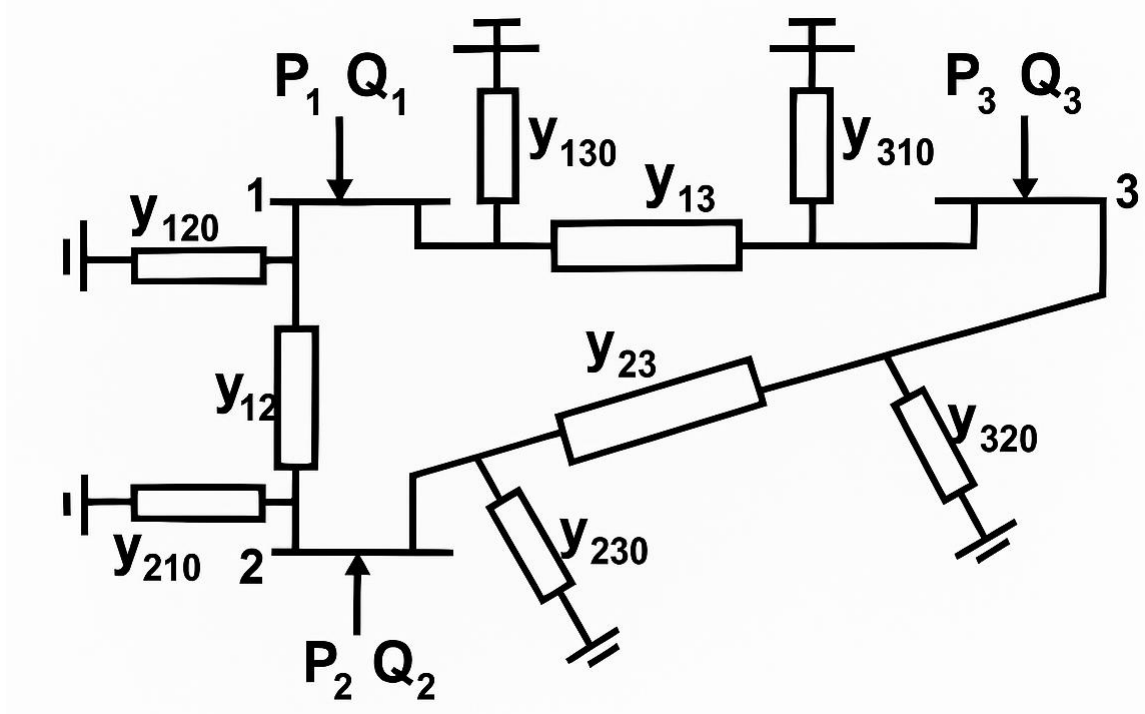
$$p_i + j q_i = \sum_{k=1}^3 v_i v_k (G_{ik} \cos(\theta_i - \theta_k) + B_{ik} \sin(\theta_i - \theta_k)) + j \sum_{k=1}^3 v_i v_k (G_{ik} \sin(\theta_i - \theta_k) - B_{ik} \cos(\theta_i - \theta_k))$$



$$\bar{v}_i = v_i e^{j\theta_i}$$

$$\bar{Y}_{ik} = G_{ik} + j B_{ik}$$

Power flow equations



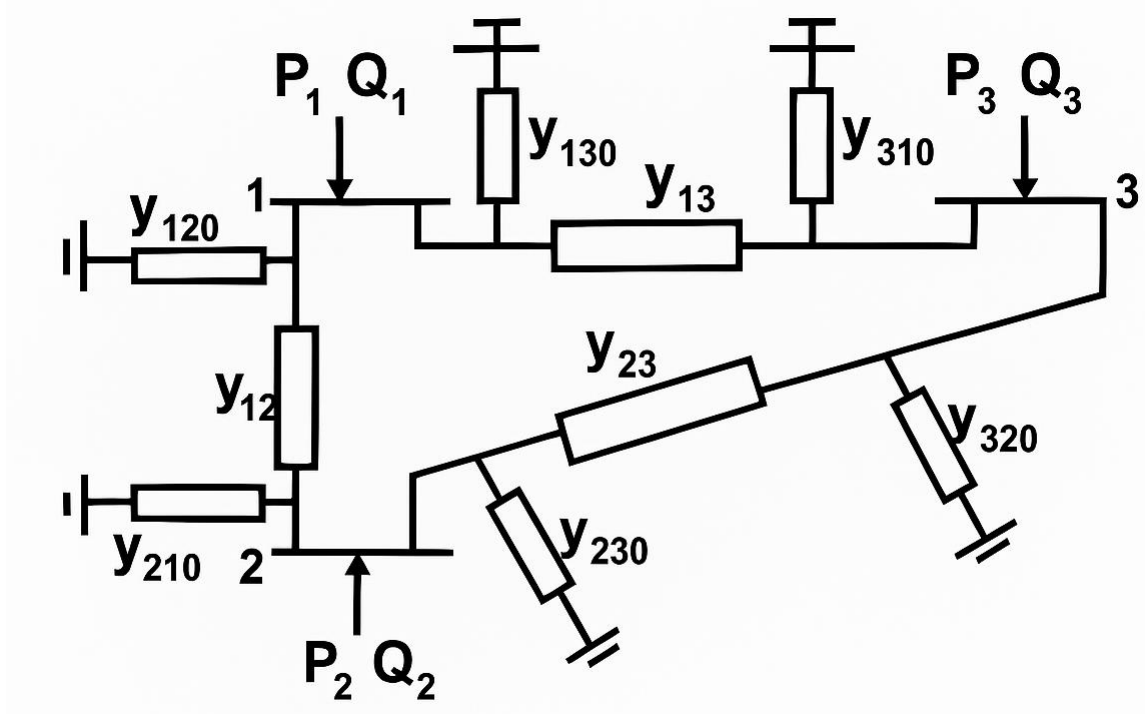
$$p_i = \sum_{k=1}^n v_i v_k (G_{ik} \cos(\theta_i - \theta_k) + B_{ik} \sin(\theta_i - \theta_k)),$$

$$q_i = \sum_{k=1}^n v_i v_k (G_{ik} \sin(\theta_i - \theta_k) - B_{ik} \cos(\theta_i - \theta_k)).$$

- Voltage (**magnitude** and **angle**)
- Power (**active** and **reactive**)

Node	# variables	# equations
1	4	2
2	4	2
3	4	2
Total	12	6

Power flow equations



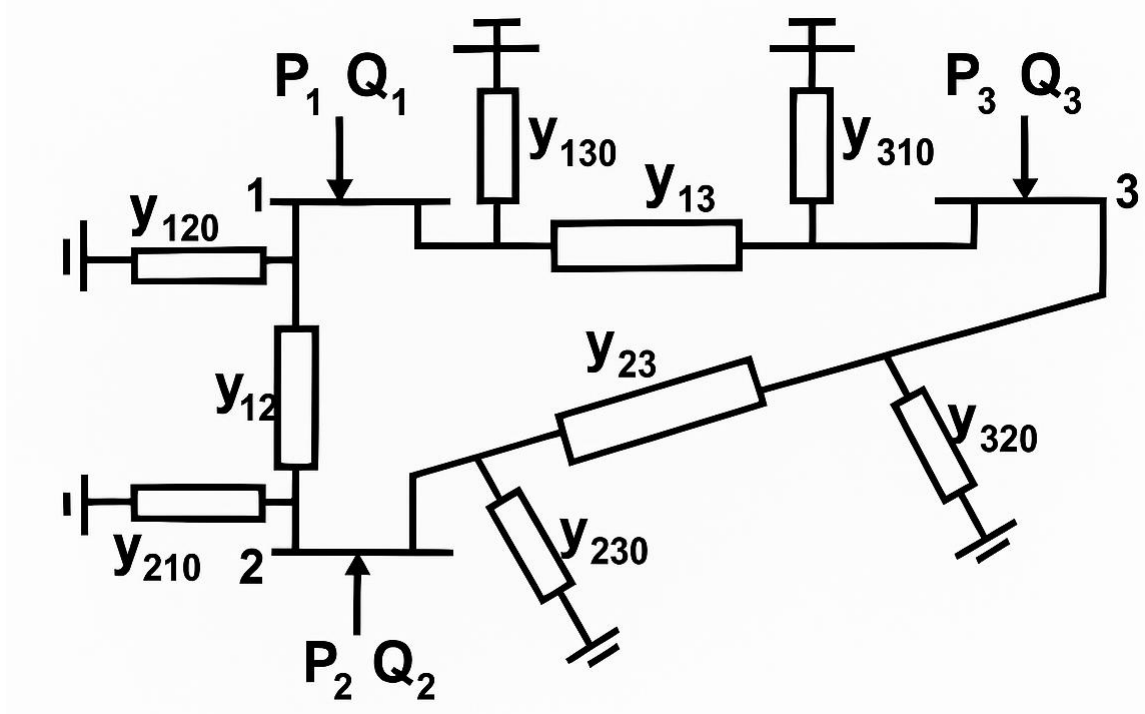
$$p_i = \sum_{k=1}^n v_i v_k (G_{ik} \cos(\theta_i - \theta_k) + B_{ik} \sin(\theta_i - \theta_k)),$$

$$q_i = \sum_{k=1}^n v_i v_k (G_{ik} \sin(\theta_i - \theta_k) - B_{ik} \cos(\theta_i - \theta_k)).$$

■ Type of nodes

Type	Known	Unknown
Slack (only one)	$V, \theta (1 0^\circ)$	P, Q
PV	P, V	Q, θ
PQ	P, Q	V, θ

Power flow equations

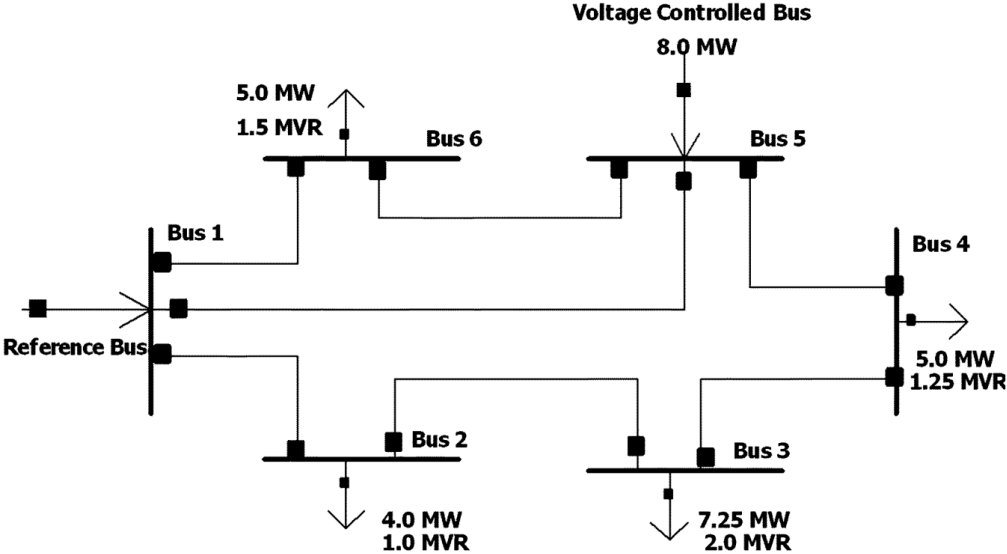


$$p_i = \sum_{k=1}^n v_i v_k (G_{ik} \cos(\theta_i - \theta_k) + B_{ik} \sin(\theta_i - \theta_k)),$$

$$q_i = \sum_{k=1}^n v_i v_k (G_{ik} \sin(\theta_i - \theta_k) - B_{ik} \cos(\theta_i - \theta_k)).$$

Node	# known variables	# unknown variables	# equations
1	2	2	2
2	2	2	2
3	2	2	2
Total		6	6

Power flow equations



$$p_i = \sum_{k=1}^n v_i v_k (G_{ik} \cos(\theta_i - \theta_k) + B_{ik} \sin(\theta_i - \theta_k)),$$
$$q_i = \sum_{k=1}^n v_i v_k (G_{ik} \sin(\theta_i - \theta_k) - B_{ik} \cos(\theta_i - \theta_k)).$$

Node	P	Q	V	θ	# unknown variables	# equations
1						
2						
3						
4						
5						
6						



Power flow: iterative approaches

- Gauss Seidel (small systems)
 - Newton-Raphson (large and meshed systems)
 - It uses the **Jacobian matrix**
 - Backward-Forward (distribution systems)
 - ...and more
-
- Software: Neplan, Digsilent, Matpower,
Pandapower



Gauss-Seidel Method

```
INPUT: Ybus, bus types, Qmin, Qmax, |V|_spec, tolV, max_iter, alpha
INITIALIZE: |V|,  $\delta$  according to bus types
for k = 0 .. max_iter-1:
    V_old = copy(V)
    for each bus i  $\neq$  Slack:
        sumYV =  $\sum_{j \neq i} Y[i,j] * V[j]$       # Uses latest available values inline
        if PQ bus:
            V_calc = ((P[i] - j*Q[i]) / conj(V[i]) - sumYV ) / Y[i,i]
            V[i] = V[i] + alpha * (V_calc - V[i])
        else if PV bus:
            Q_calc[i] = -imag( conj(V[i]) * (sumYV + Y[i,i]*V[i]) )
            if Q_calc[i] > Qmax[i] or Q_calc[i] < Qmin[i]:
                Q_eff = max(Qmin[i], min(Q_calc[i], Qmax[i]))
                V_calc = ((P[i] - j*Q_eff) / conj(V[i]) - sumYV ) / Y[i,i]
                V[i] = V[i] + alpha * (V_calc - V[i])      # Treats as PQ bus
            else:
                V_calc = ((P[i] - j*Q_calc[i]) / conj(V[i]) - sumYV ) / Y[i,i]
                V_acc = V[i] + alpha * (V_calc - V[i])
                V[i] = |V|_spec[i] * exp(j*angle(V_acc)) # Update angle only
    if max(|V - V_old|)  $\leq$  tolV: return V (bus voltages)
```


Exercise :

Base values:

- 25 kV
- 100 MVA

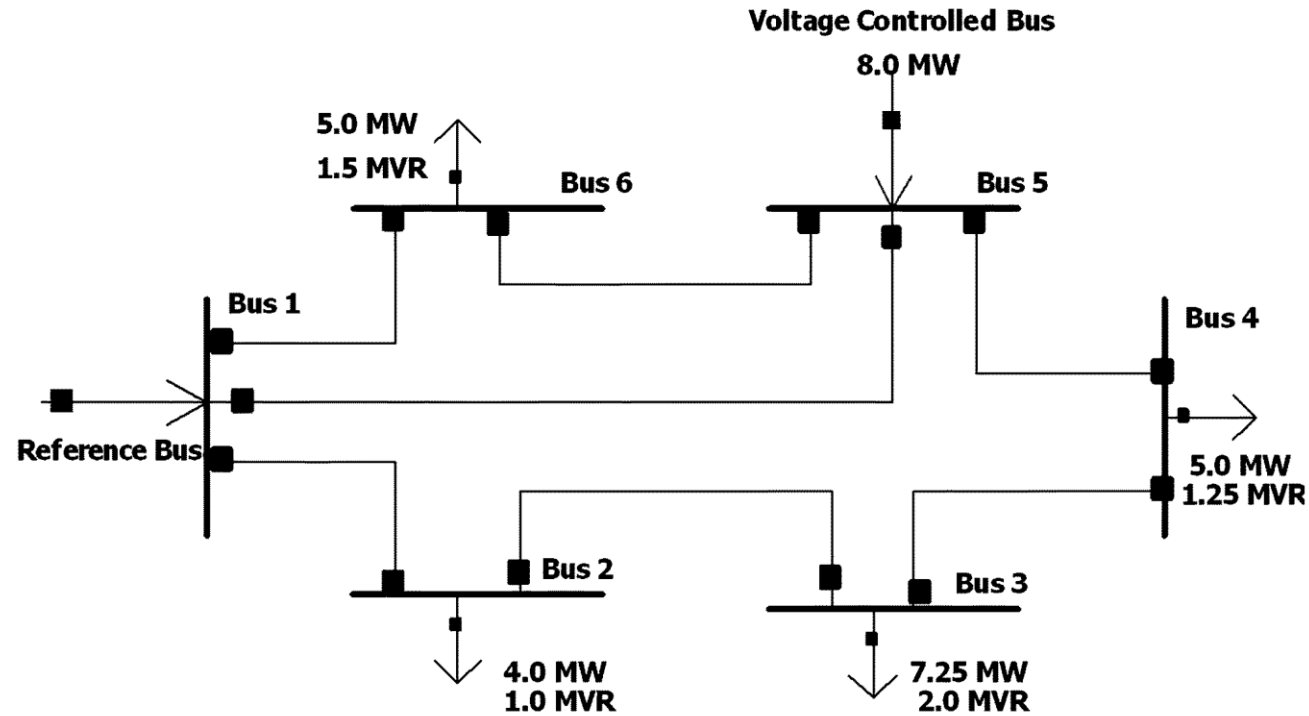


Table 1: Bus Data

Bus No.	Bus Type	Voltage (p.u.)		Bus Power	
		$ V $	δ	(MW)	(MVAR)
1	SB	1	0	Slack bus	
2	PQ	—	—	-4	-1
3	PQ	—	—	-7.25	-2
4	PQ	—	—	-5	-1.25
5	PV	1	—	8	—
6	PQ	—	—	-5	-1.5

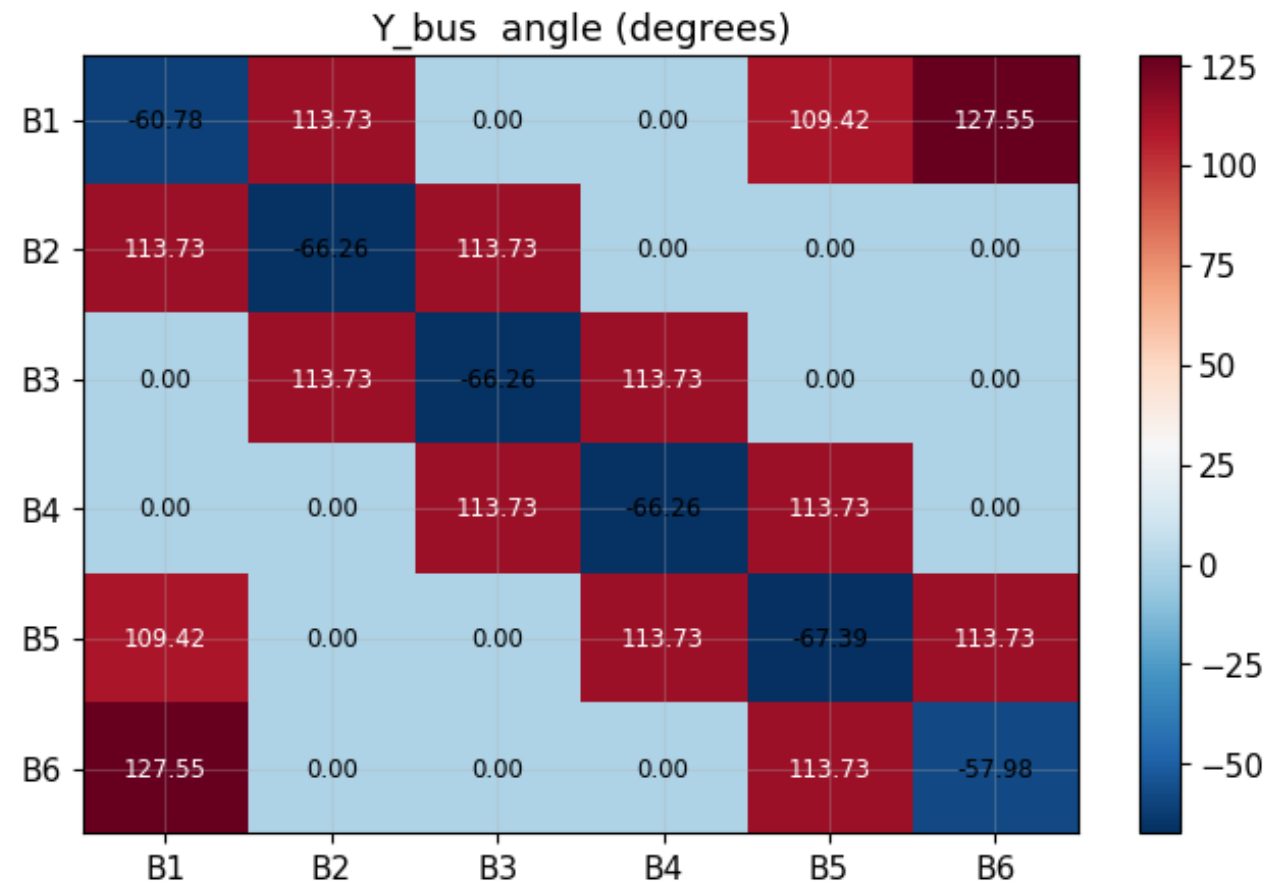
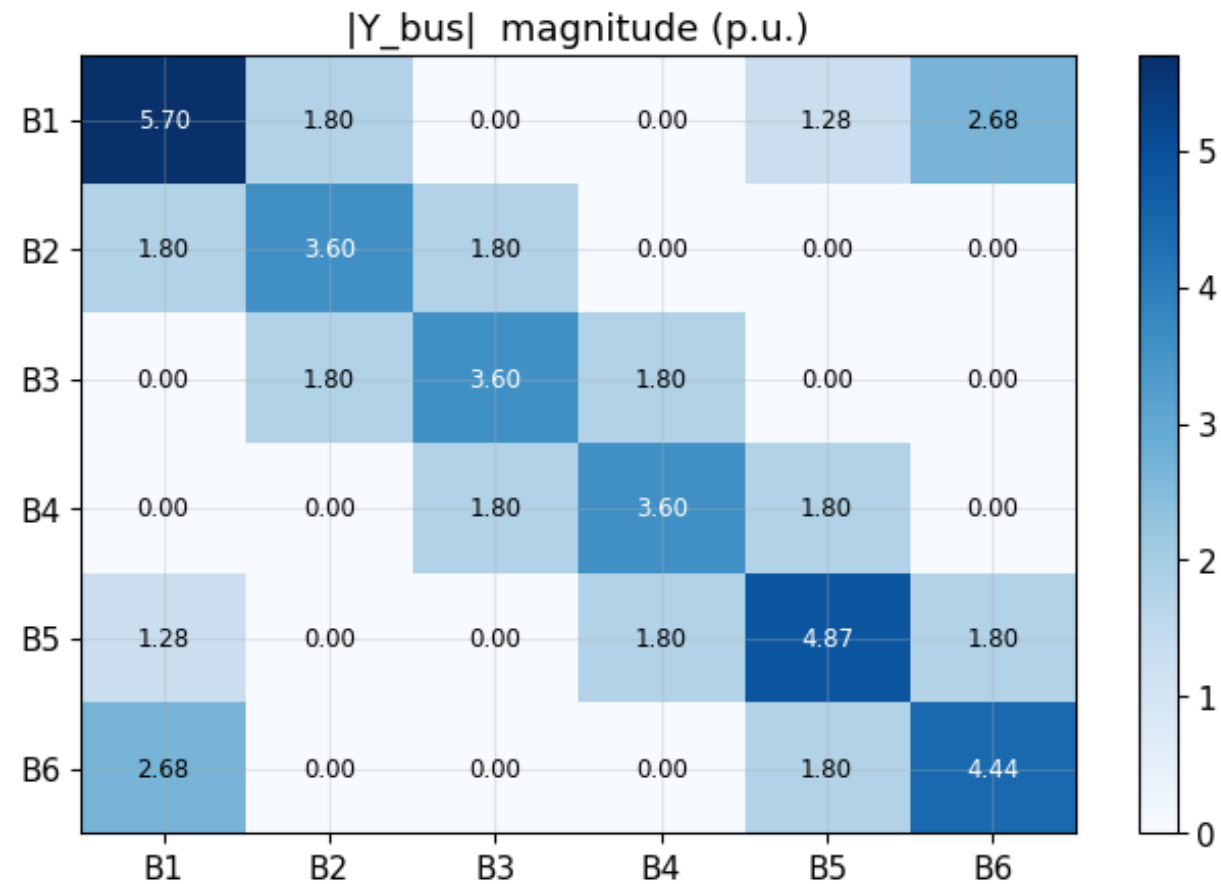
Table 2: Line Data

From Bus	To Bus	R_{line} (p.u.)	X_{line} (p.u.)	Y_{shunt} (p.u.)
1	2	0.2238	0.5090	0.0012
2	3	0.2238	0.5090	0.0012
3	4	0.2238	0.5090	0.0012
4	5	0.2238	0.5090	0.0012
5	6	0.2238	0.5090	0.0012
6	1	0.2276	0.2961	0.0025
1	5	0.2603	0.7382	0.0008



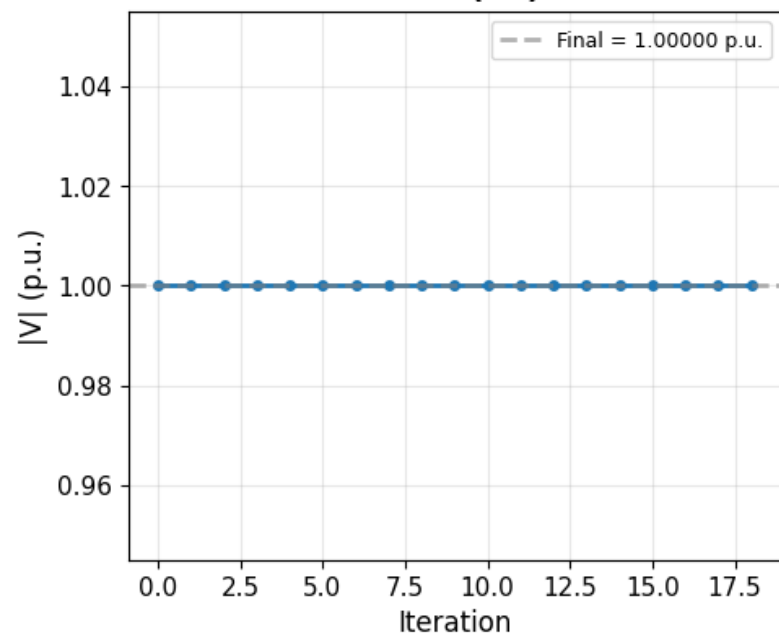
Admittance matrix

Bus Admittance Matrix Y_{bus}

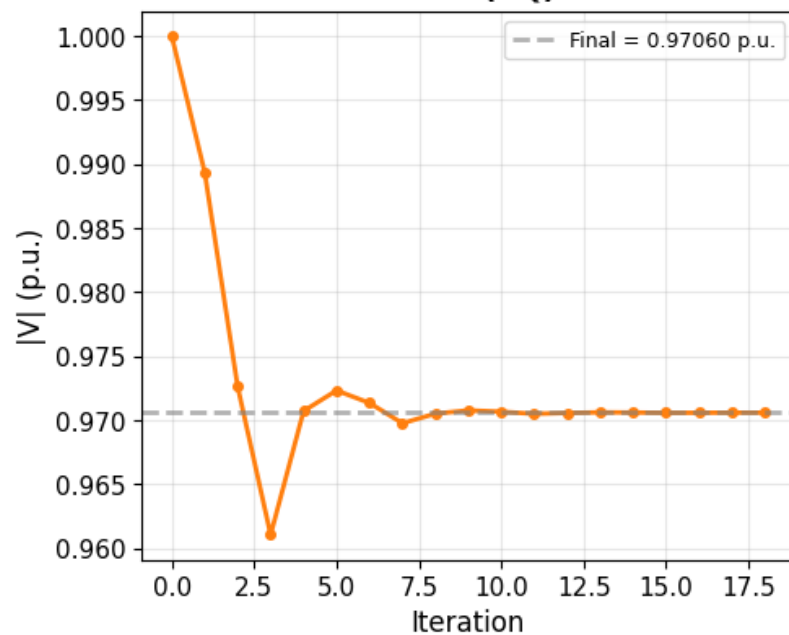


Voltage Magnitude per Bus - Gauss-Seidel Convergence

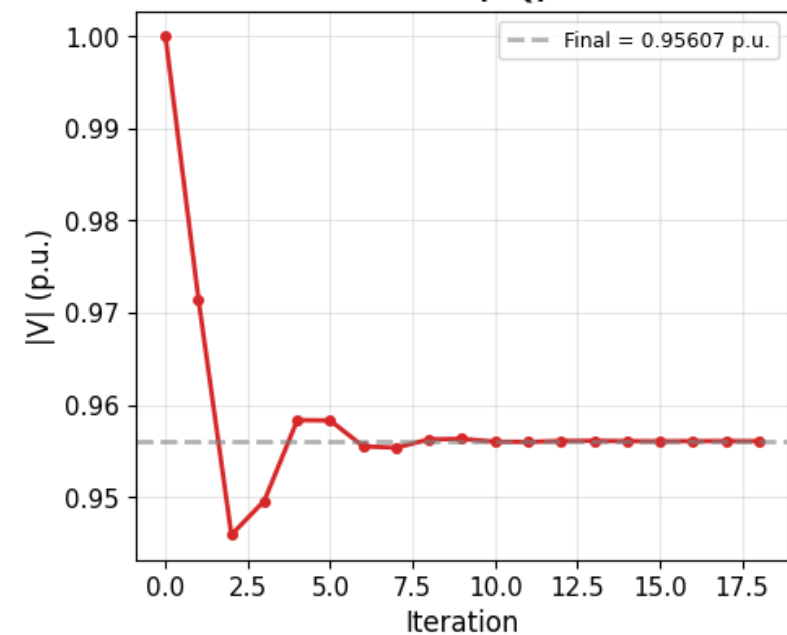
Bus 1 (SB)



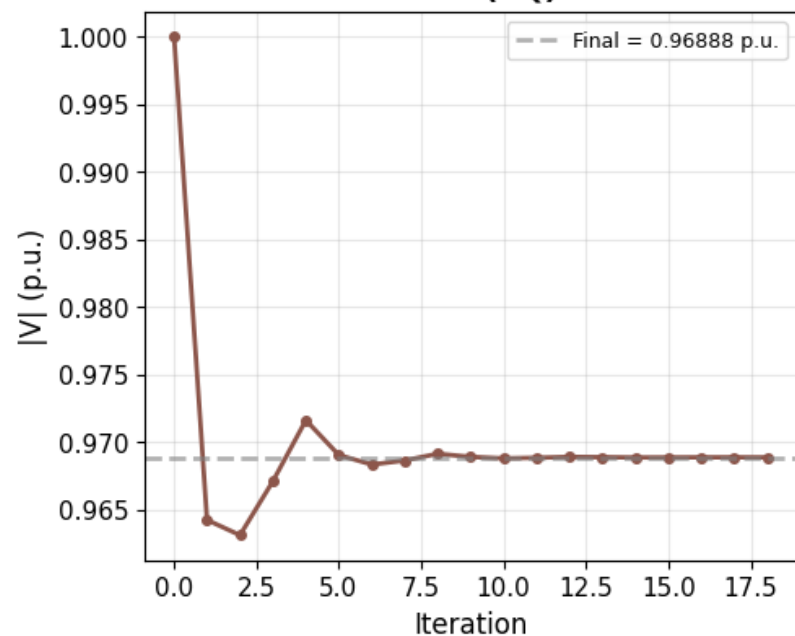
Bus 2 (PQ)



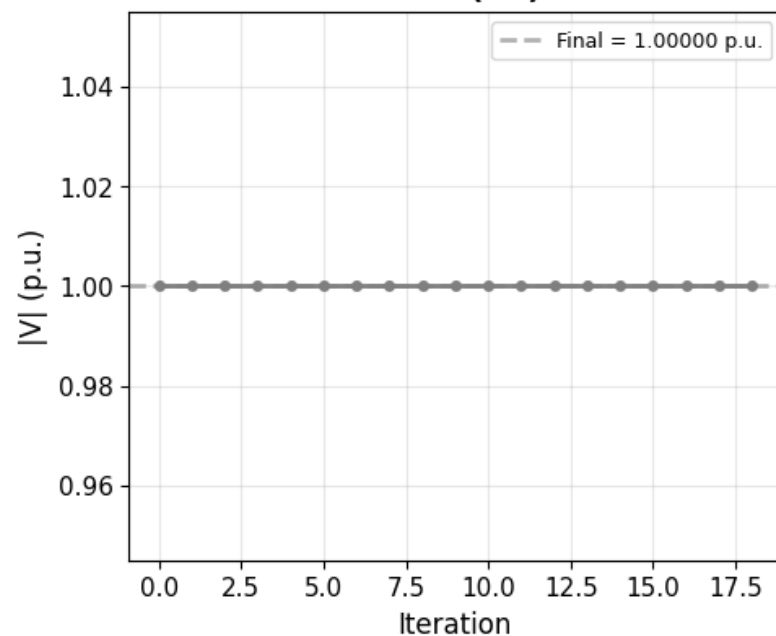
Bus 3 (PQ)



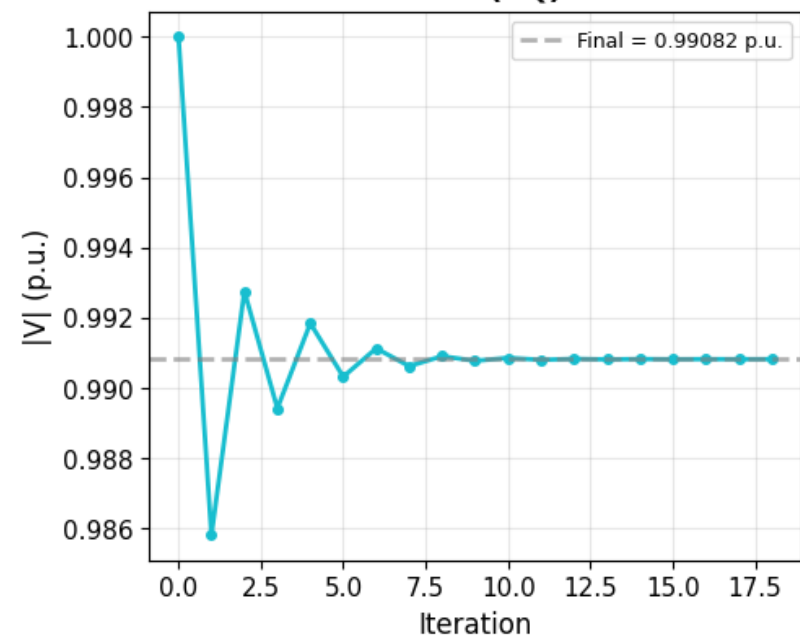
Bus 4 (PQ)

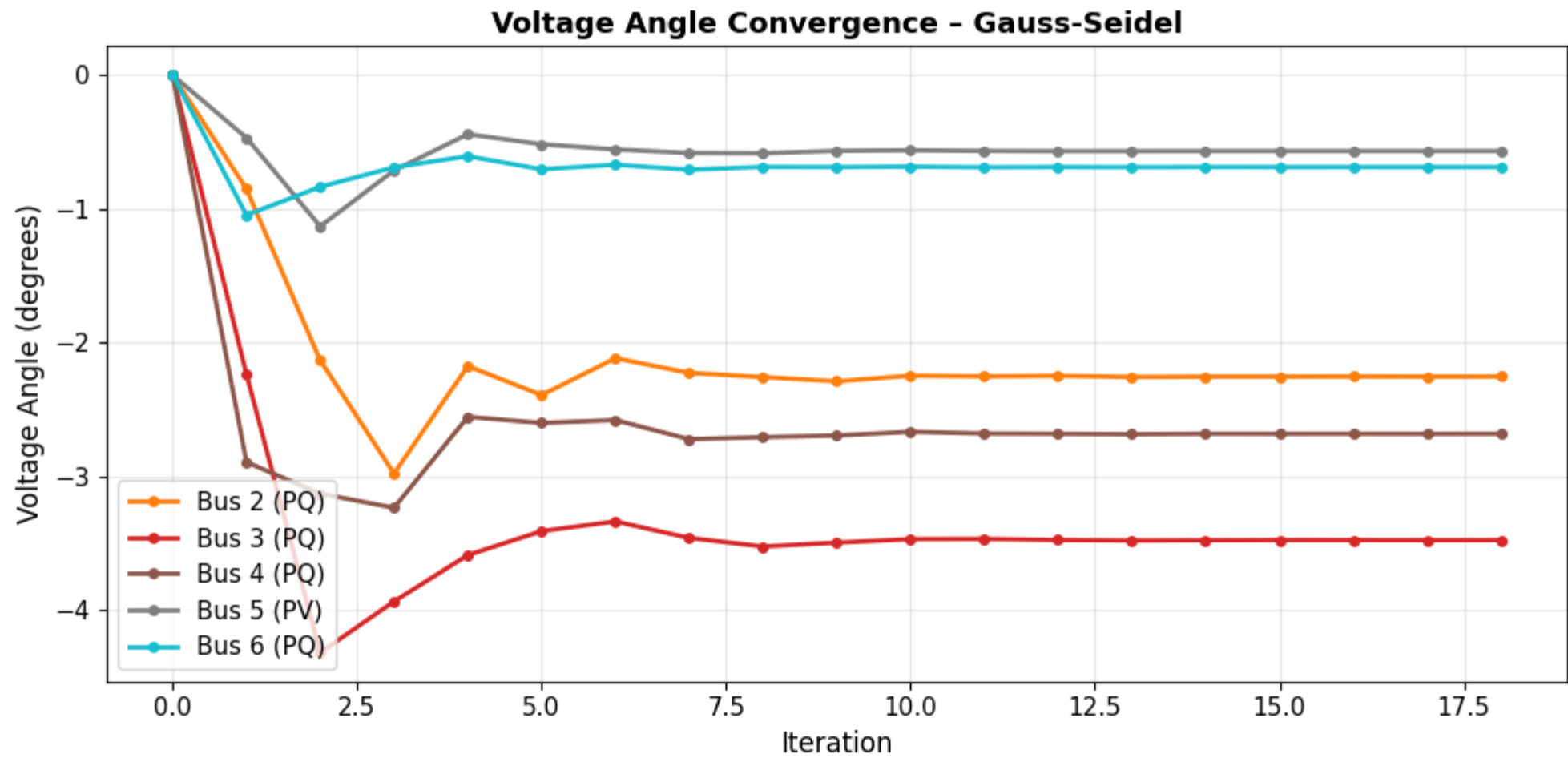


Bus 5 (PV)

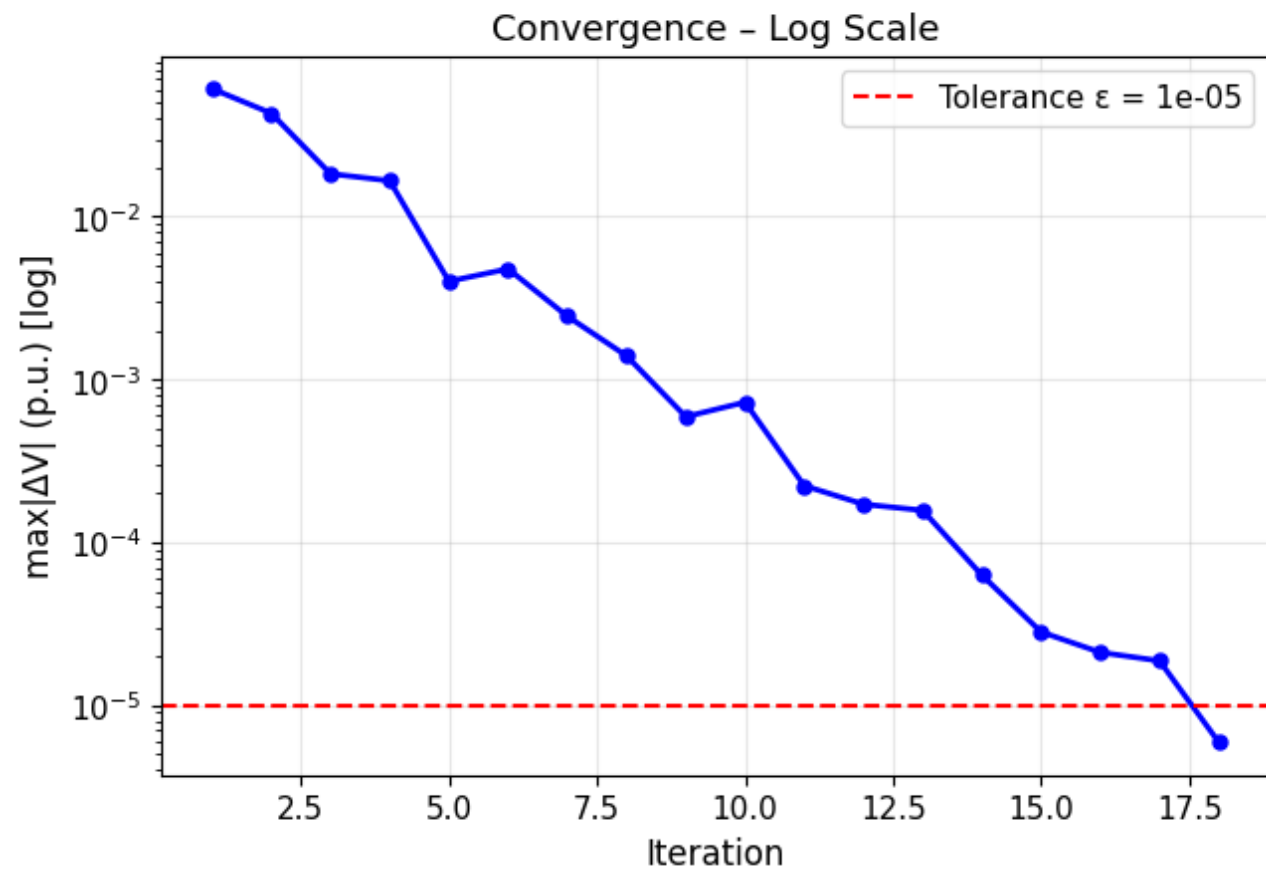
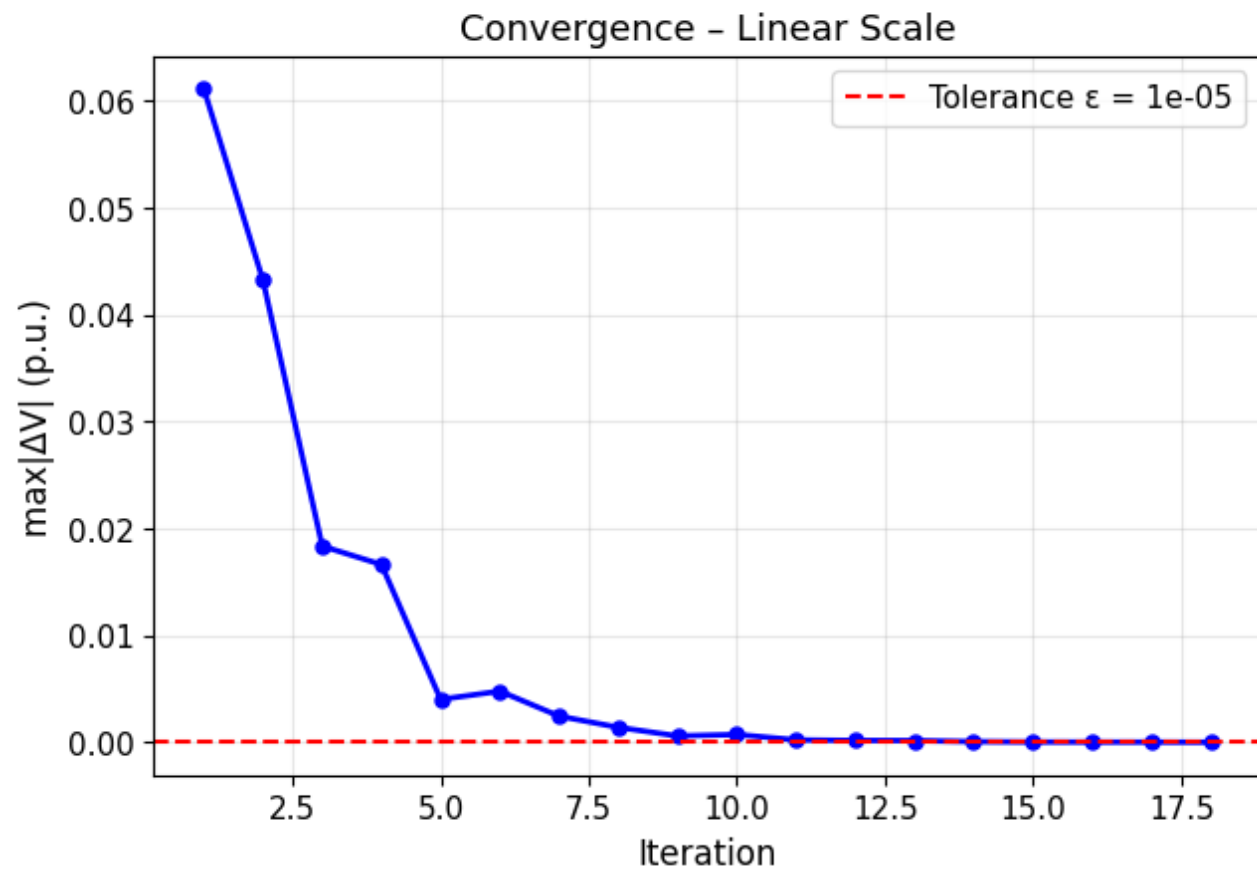


Bus 6 (PQ)

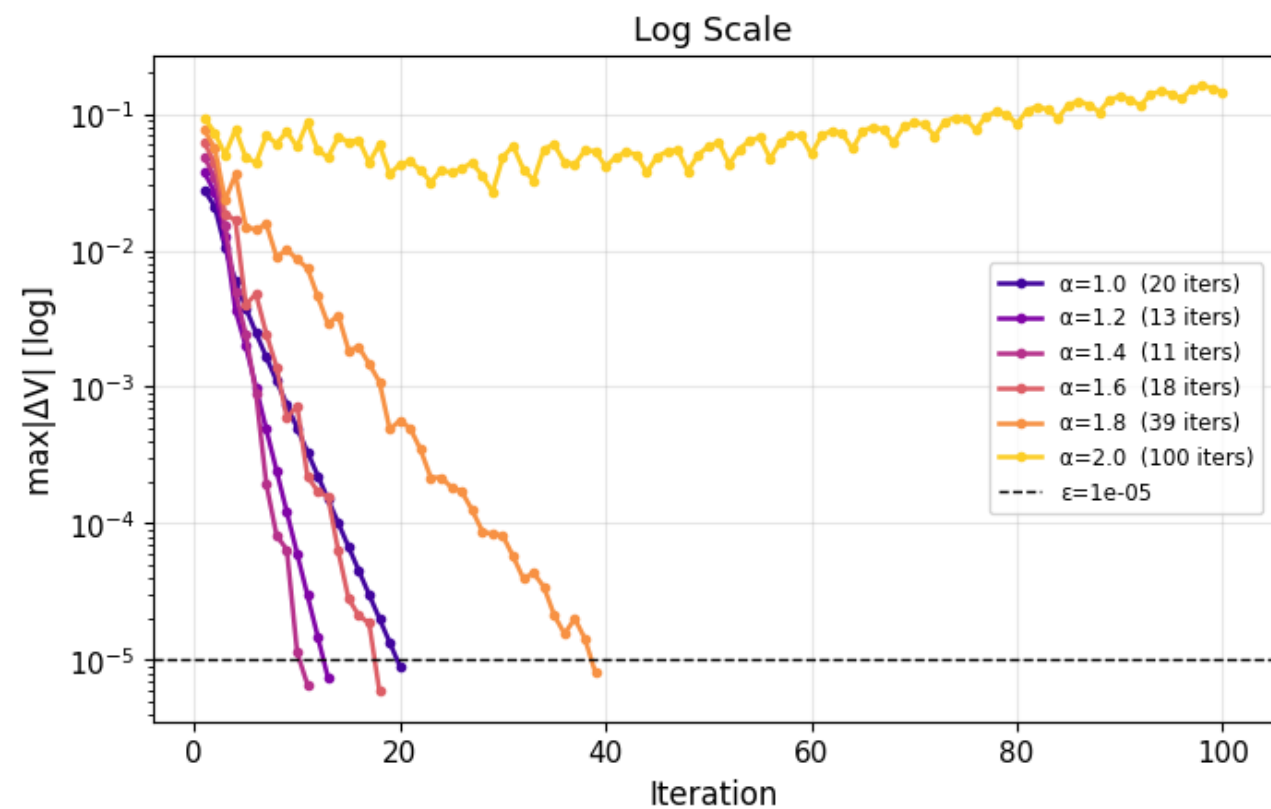
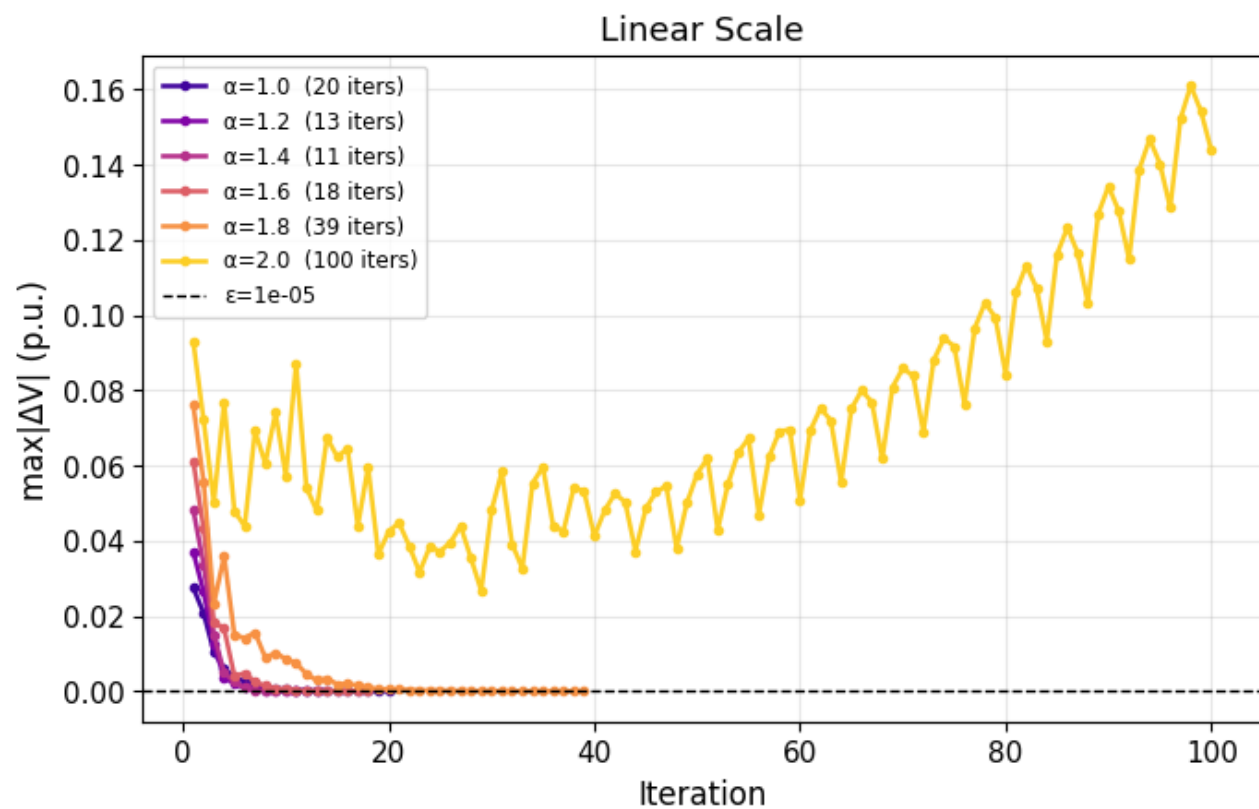




Gauss-Seidel Convergence History



Effect of Acceleration Factor α



Known issues of Gauss-Seidel method

- Slow converge rate.
- Higher grid size, higher computation time.
- Converge issues (highly loaded, radial configuration).



Jacobian matrix

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} J \end{bmatrix} \begin{bmatrix} \Delta \theta \\ \Delta V \end{bmatrix}$$

$$D_{\delta} \mathbf{S} = j \operatorname{diag}(\mathbf{V}) [\operatorname{diag}(\mathbf{Y}^* \mathbf{V}^*) - \mathbf{Y}^* \operatorname{diag}(\mathbf{V}^*)]$$

$$D_{|\mathbf{V}|} \mathbf{S} = \operatorname{diag}(\mathbf{V}) [\operatorname{diag}(\mathbf{Y}^* \mathbf{V}^*) + \mathbf{Y}^* \operatorname{diag}(\mathbf{V}^*)] \operatorname{diag}(|\mathbf{V}|^{-1})$$

$$\mathbf{J} = \begin{pmatrix} \mathbf{J}_{11} & \mathbf{J}_{12} \\ \mathbf{J}_{21} & \mathbf{J}_{22} \end{pmatrix}_{2N \times 2N},$$

$$\mathbf{J}_{11} = \Re(D_{\delta} \Delta \mathbf{S}), \quad \mathbf{J}_{12} = \Re(D_{|\mathbf{V}|} \Delta \mathbf{S}),$$

$$\mathbf{J}_{21} = \Im(D_{\delta} \Delta \mathbf{S}), \quad \mathbf{J}_{22} = \Im(D_{|\mathbf{V}|} \Delta \mathbf{S}).$$

Newton-Raphson Method

INPUT: Ybus, bus types, tolP, tolQ, max_iter

INITIALIZE: $|V|$, δ according to bus types

for k = 0..max_iter-1:

 compute $V_i = |V_i| \angle \delta_i$ for all i

 for each bus i: compute P_i , Q_i from V and Ybus

 form $\Delta P_i = P_{ref_i} - P_i$ for all non-slack buses

 form $\Delta Q_i = Q_{ref_i} - Q_i$ for PQ buses

if $\max(|\Delta P|) \leq \text{tolP}$ and $\max(|\Delta Q|) \leq \text{tolQ}$: **return** converged solution

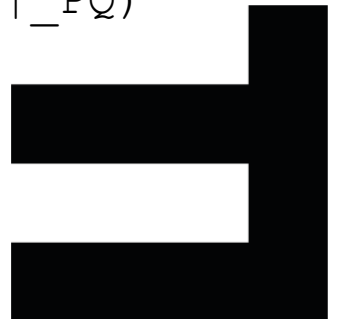
 compute Jacobian using current V, δ , Ybus

 form Jacobian J (rows: $\Delta P_{\text{non-slack}}$, ΔQ_{PQ} ; cols: $\Delta \delta_{\text{non-slack}}$, $\Delta |V|_{\text{PQ}}$)

 solve $\mathbf{J} * \mathbf{x} = [\Delta \mathbf{P}; \Delta \mathbf{Q}]$ for $\mathbf{x} = [\Delta \delta; \Delta |V|]$

update $\delta_{\text{non-slack}} += \Delta \delta$

update $|V|_{\text{PQ}} += \Delta |V|$



Newton-Raphson Method

```
INPUT: Ybus, bus types, P_sch, Q_sch, |V|_spec, Qmin, Qmax, tolP, tolQ, max_iter
INITIALIZE: |V|,  $\delta$  according to bus types
for k = 0 .. max_iter-1:
    compute V[i] = |V|[i] * exp(j *  $\delta$ [i]) for all buses i
    for each bus i  $\neq$  Slack:
        P_calc[i] = real( V[i] * conj(  $\sum_{j} Y[i,j] * V[j]$  ) )
         $\Delta P[i]$  = P_sch[i] - P_calc[i]
        if PQ bus:
            Q_calc[i] = -imag( V[i] * conj(  $\sum_{j} Y[i,j] * V[j]$  ) )
             $\Delta Q[i]$  = Q_sch[i] - Q_calc[i]
    if max(| $\Delta P$ |)  $\leq$  tolP and max(| $\Delta Q$ |)  $\leq$  tolQ:
        return V (converged solution)
    compute Jacobian using current V,  $\delta$ , Ybus
    form Jacobian J (for PV buses, omit the rows/columns corresponding to  $\Delta Q$  and  $\Delta|V|$ )
    solve J * [ $\Delta\delta$ ;  $\Delta|V|$ ] = [ $\Delta P$ ;  $\Delta Q$ ] for corrections  $\Delta\delta$  and  $\Delta|V|$ 
    update  $\delta_{\text{non-slack}} += \Delta\delta$ 
    update |V|_PQ +=  $\Delta|V|$ 
    # Enforce Generator Reactive Power Limits (PV Buses)
    for each PV bus i:
        Q_gen = -imag( V[i] * conj(  $\sum_{j} Y[i,j] * V[j]$  ) )
        if Q_gen > Qmax[i] or Q_gen < Qmin[i]:
            Q_sch[i] = max(Qmin[i], min(Q_gen, Qmax[i]))
            change bus i type to PQ for subsequent iterations
```

Exercise :

- 25 kV
- 100MVA

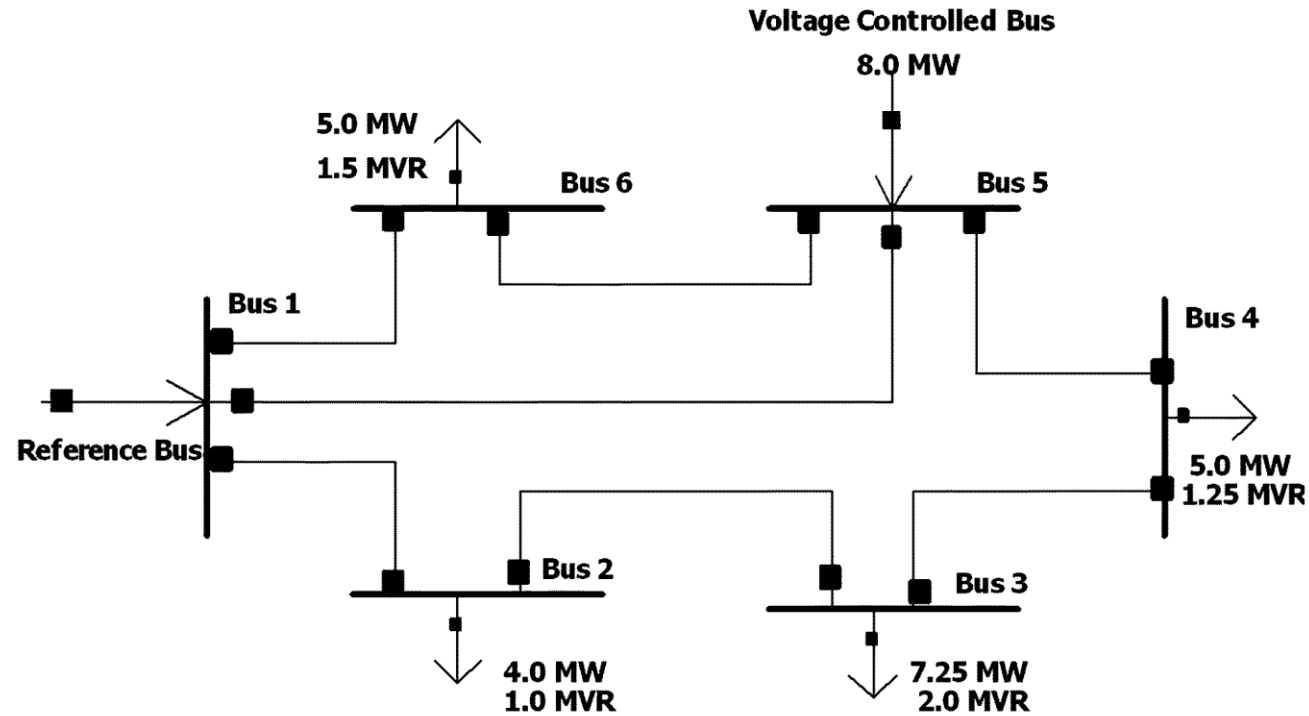


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5	PV	1	—	8	—
6	PQ	—	—	-5	-1.5

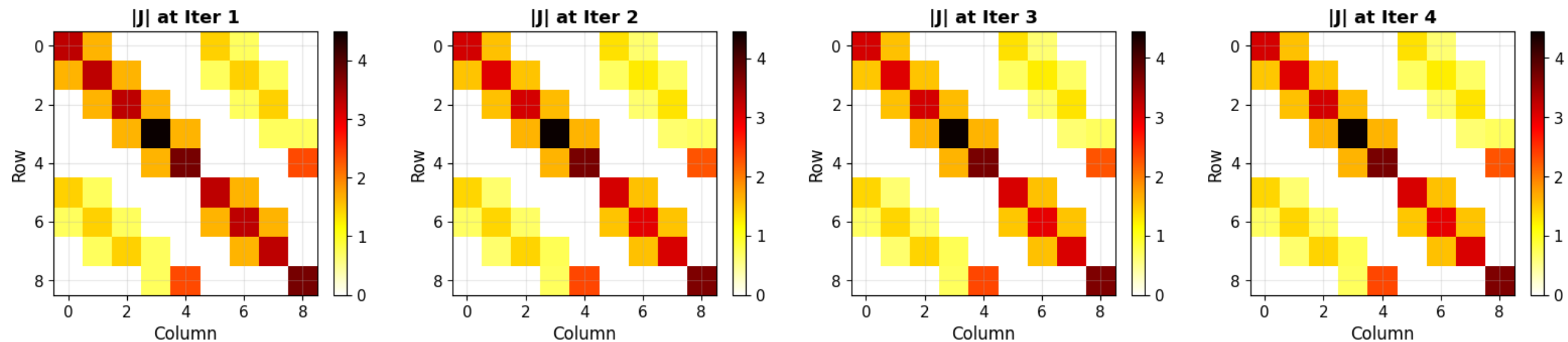
Table 2: Line Data

From Bus	To Bus	R_{line} (p.u.)	X_{line} (p.u.)	Y_{shunt} (p.u.)
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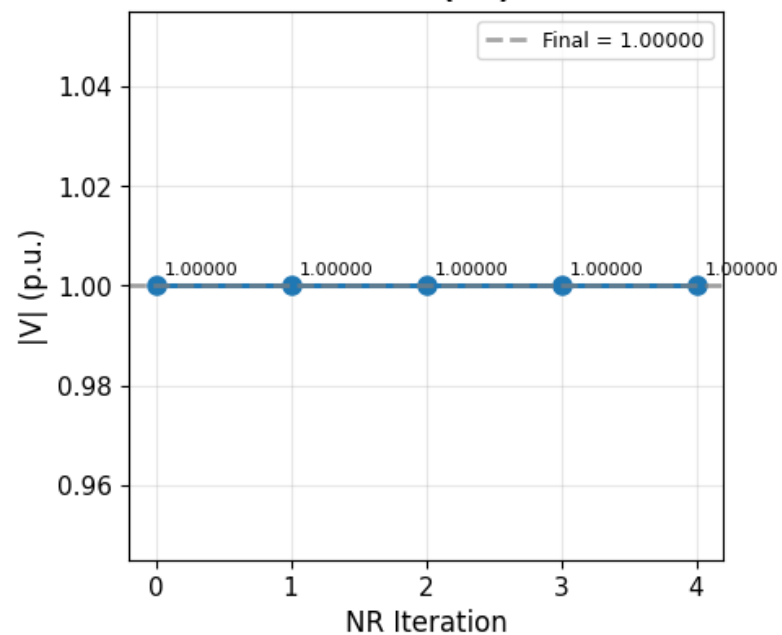
Jacobian matrix

Jacobian Magnitude Heatmaps per Iteration

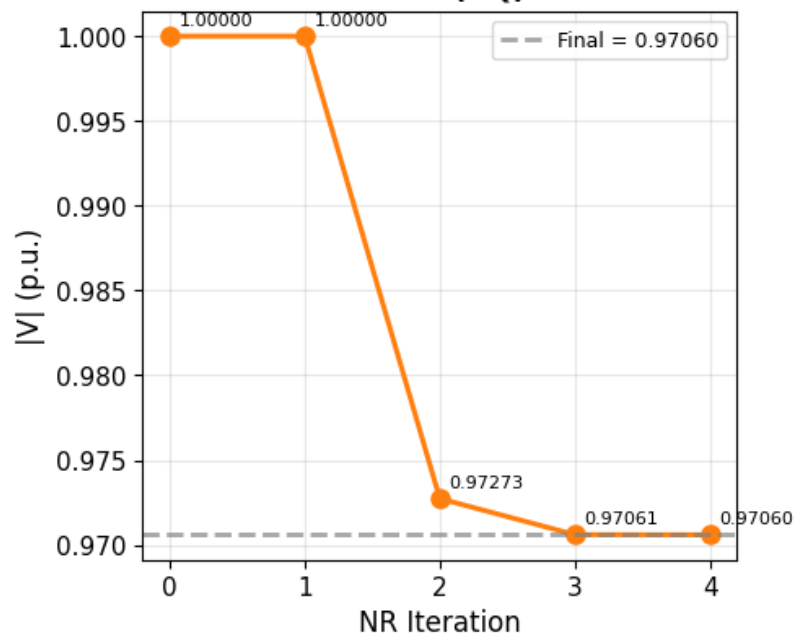


Voltage Magnitudes - NR Iterations

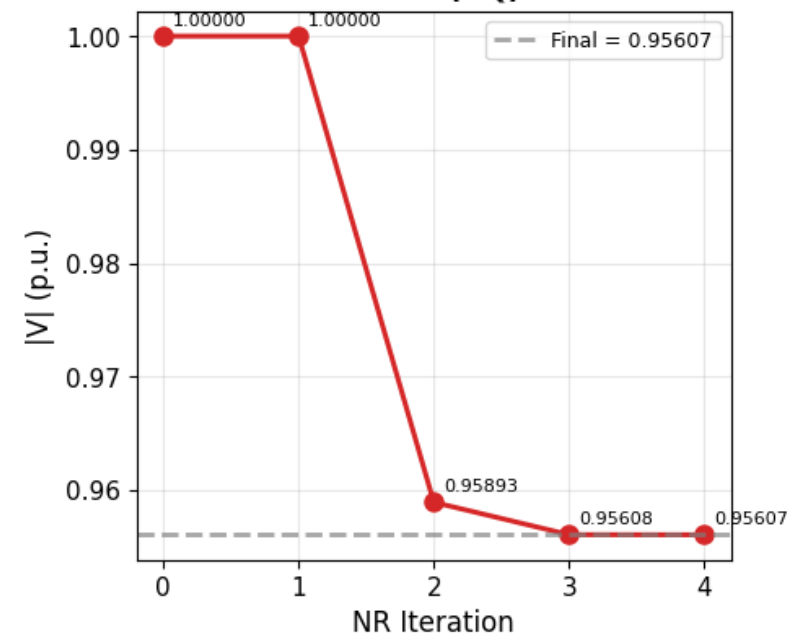
Bus 1 (SB)



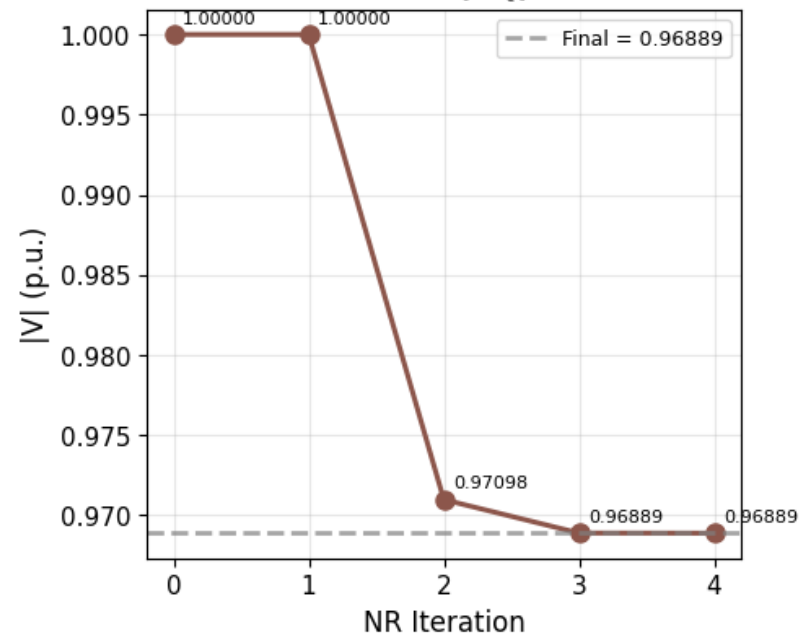
Bus 2 (PQ)



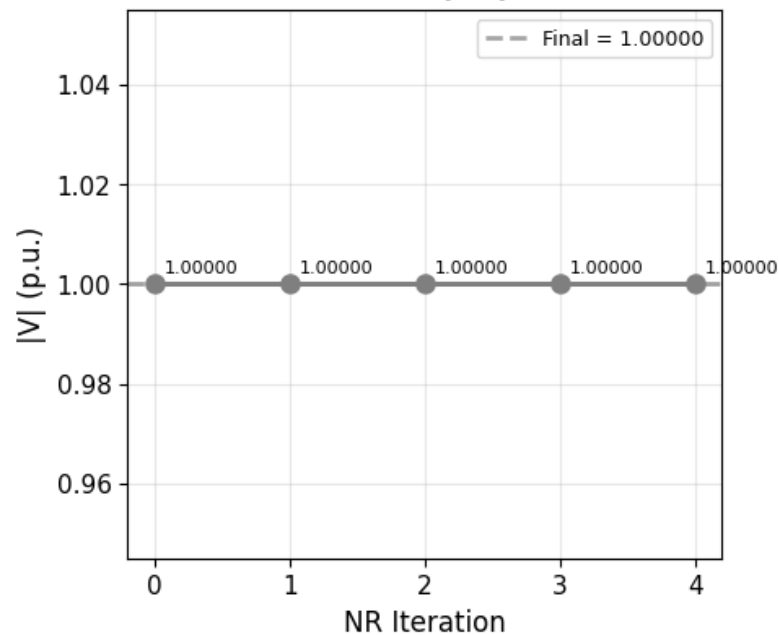
Bus 3 (PQ)



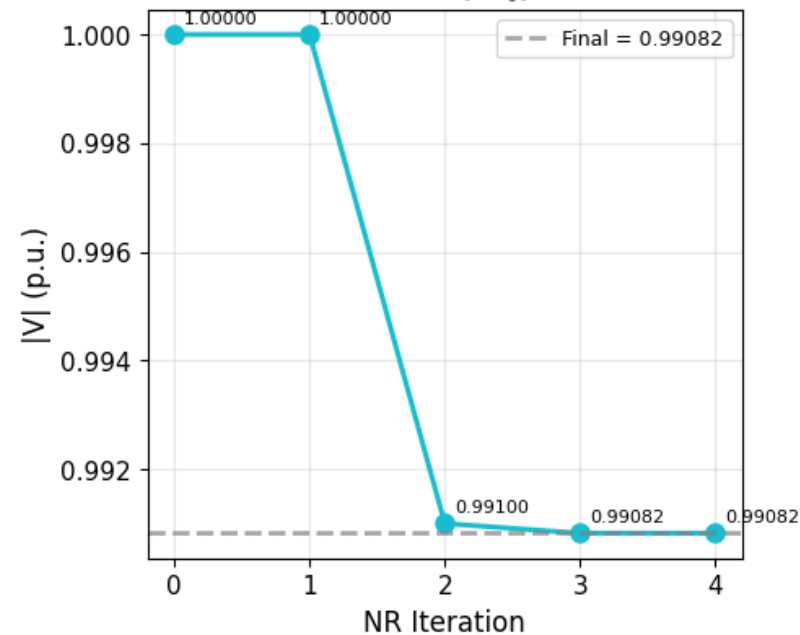
Bus 4 (PQ)



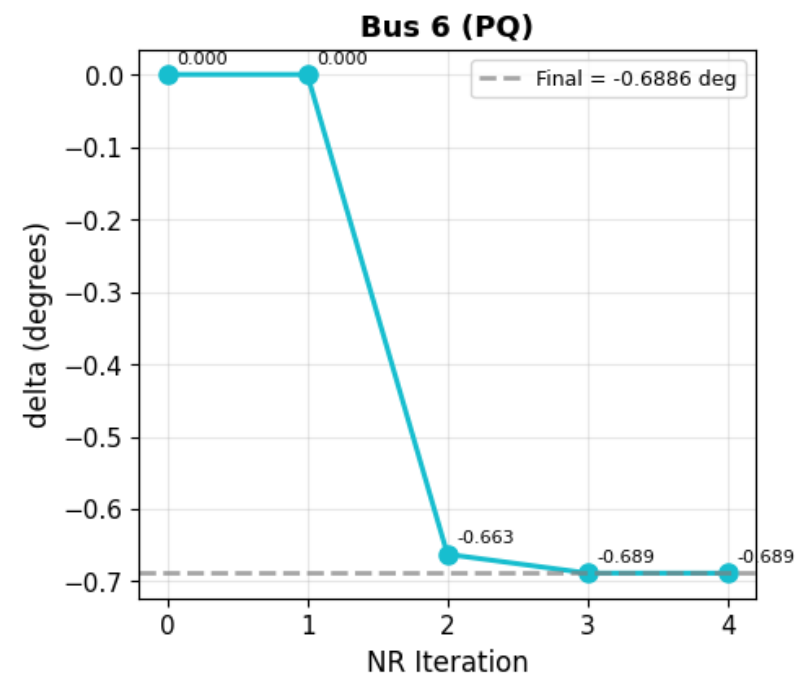
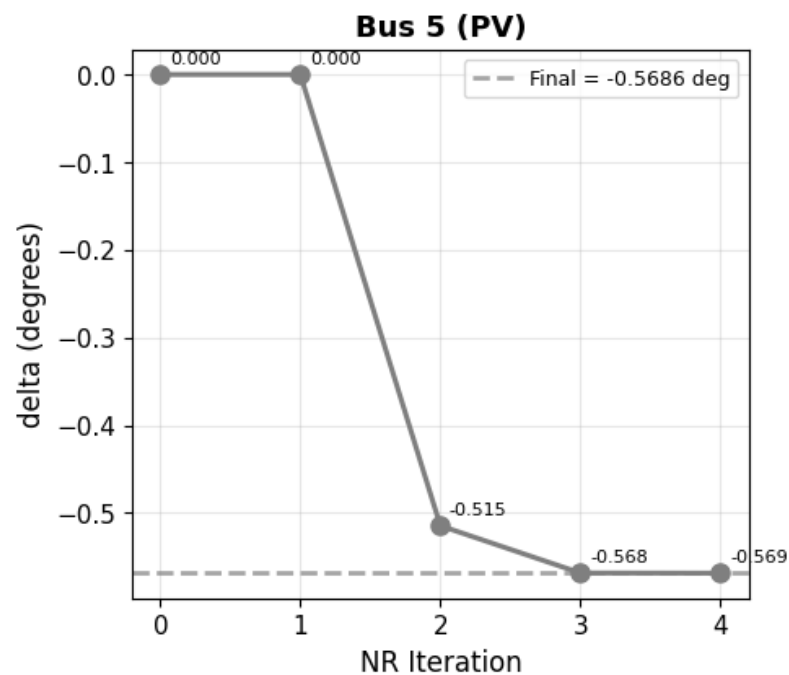
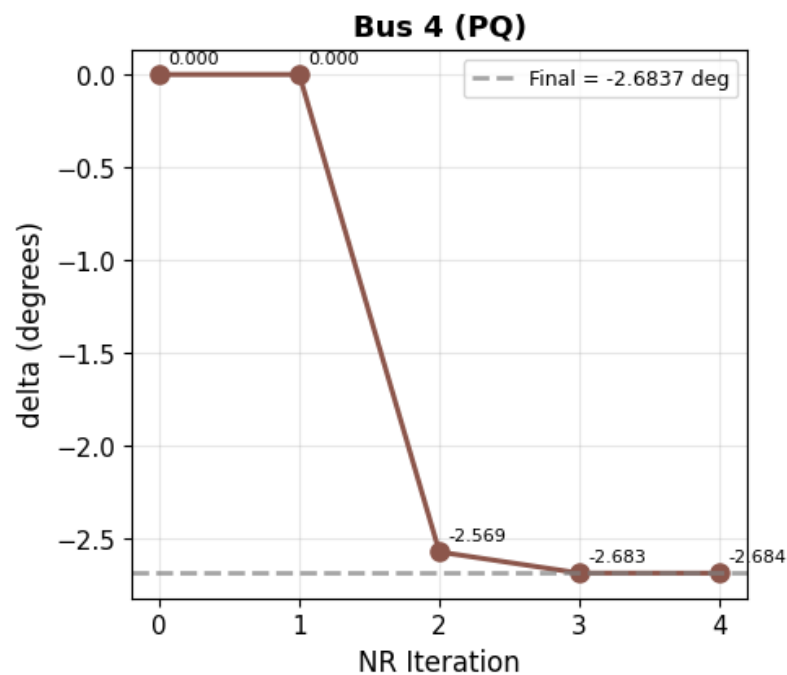
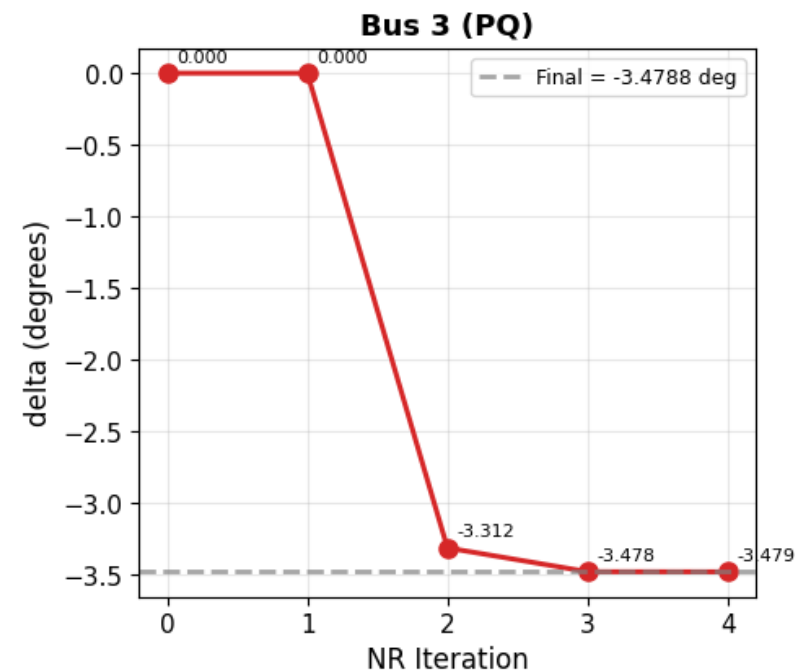
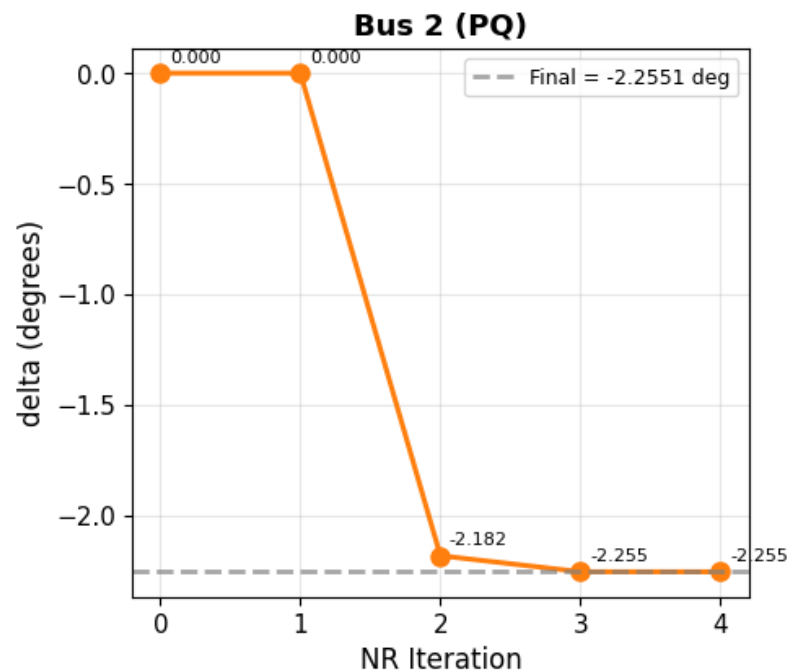
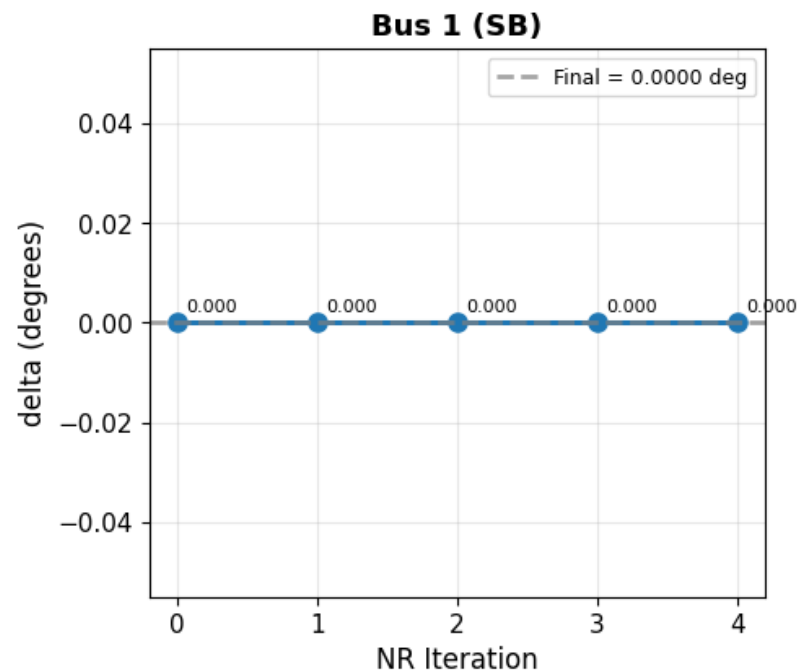
Bus 5 (PV)



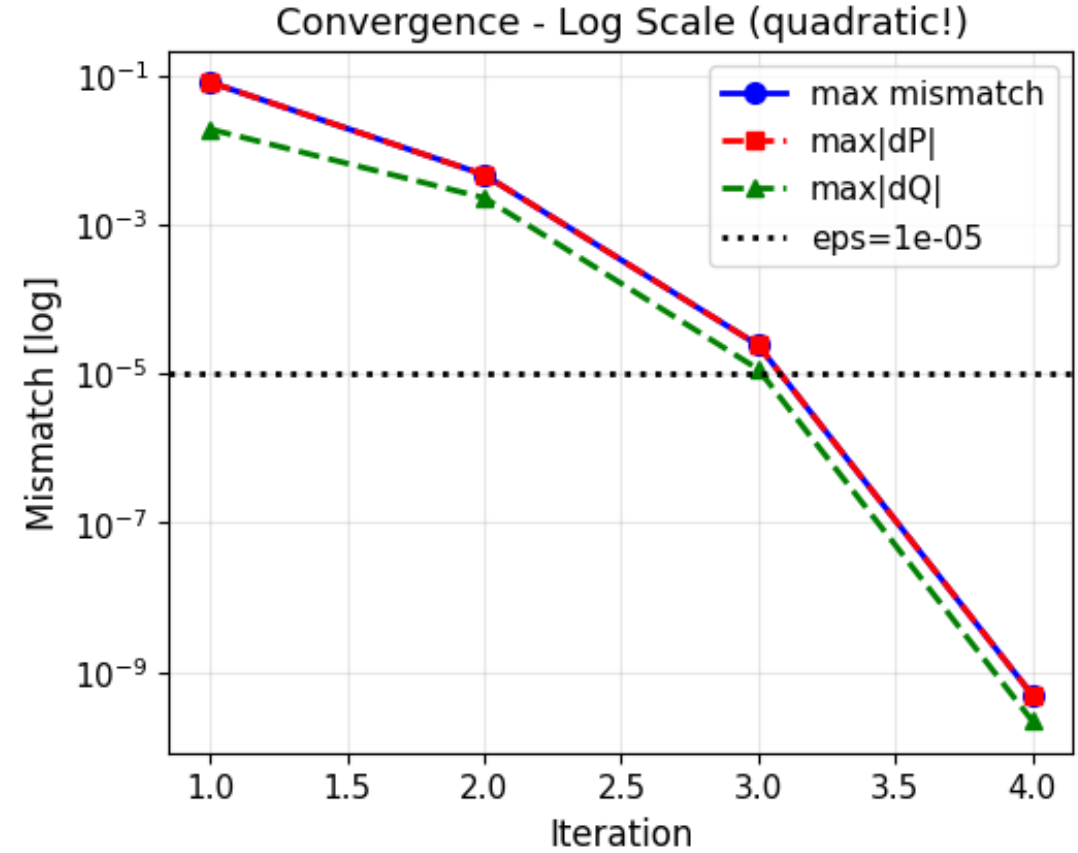
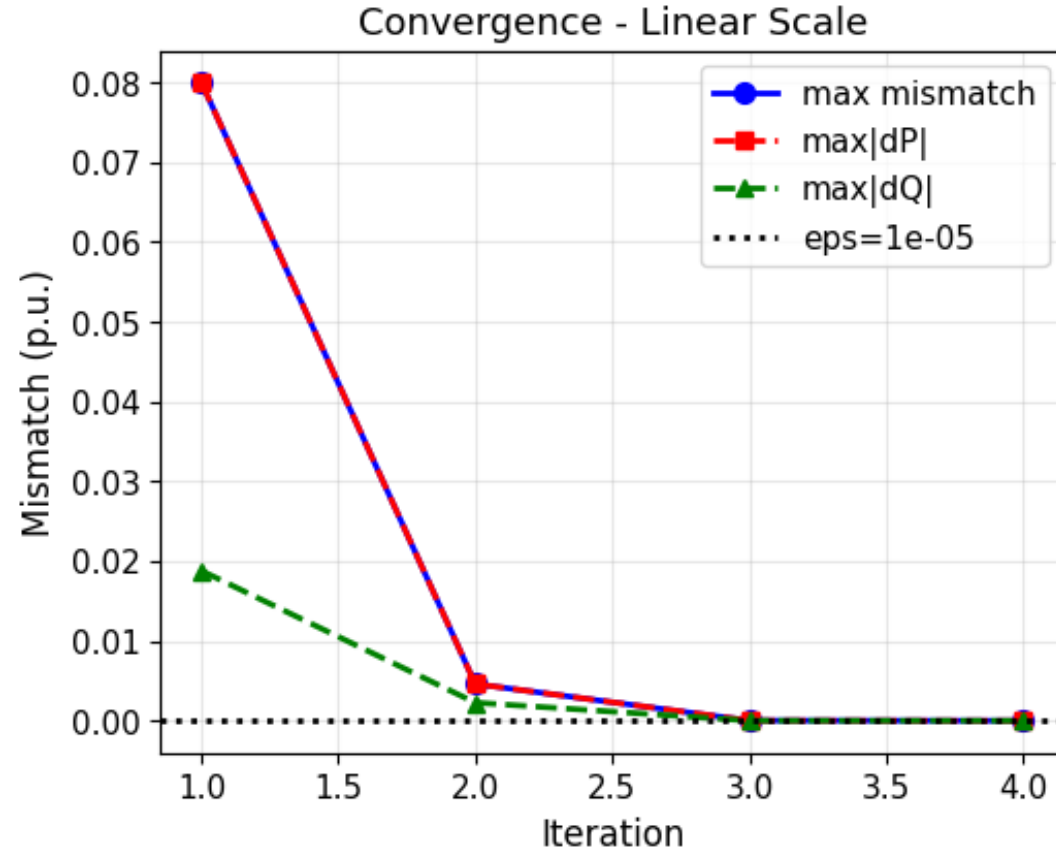
Bus 6 (PQ)



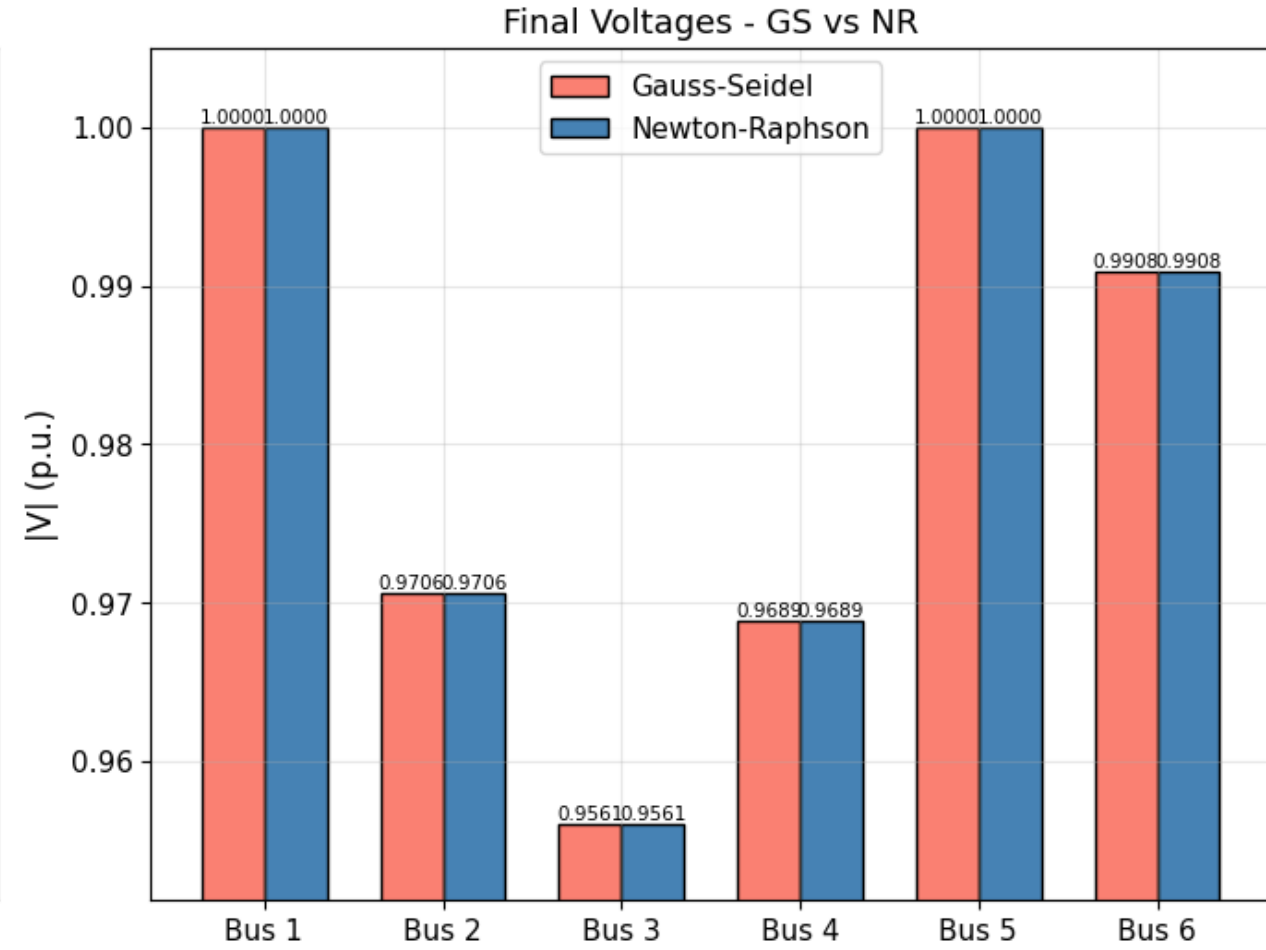
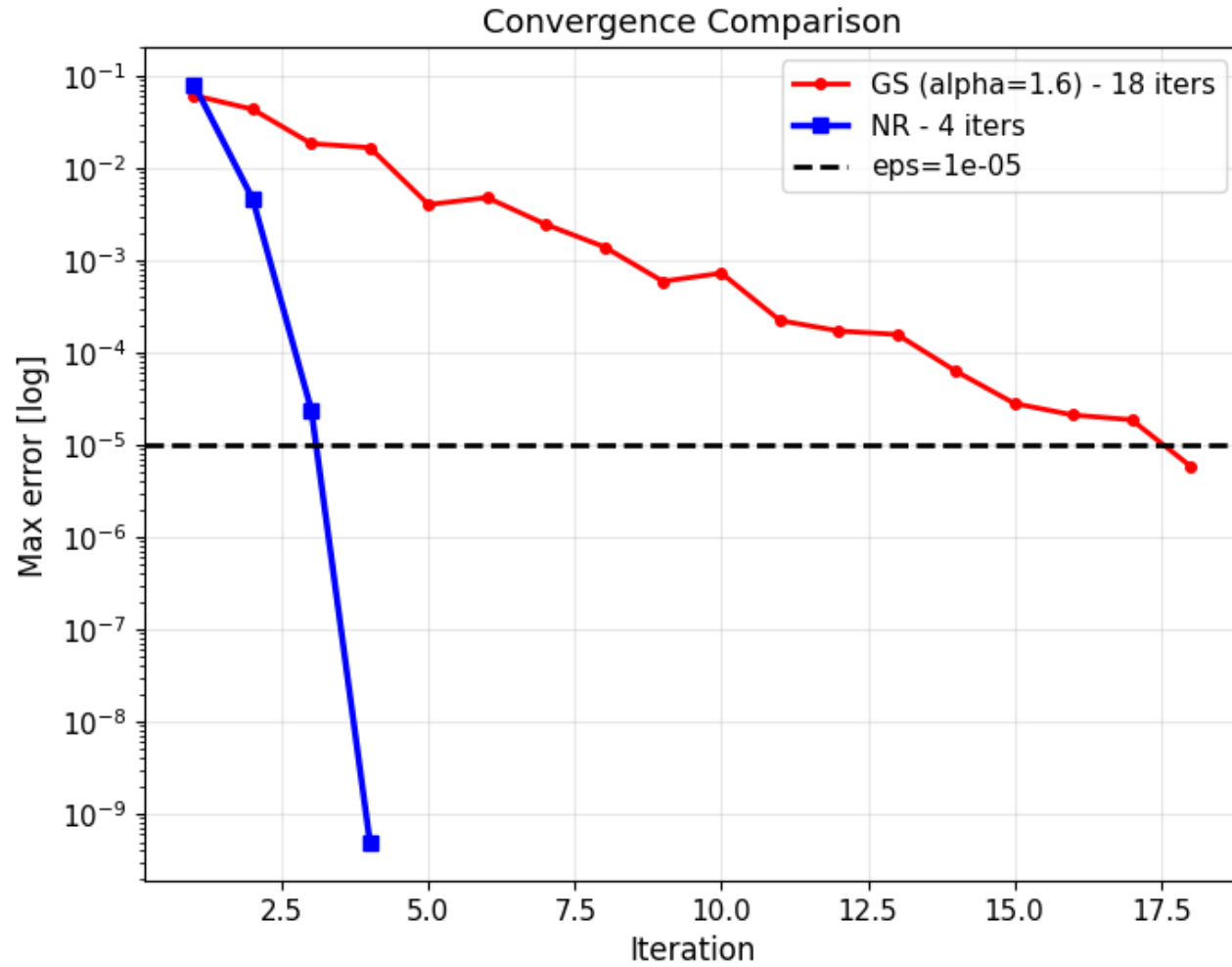
Voltage Angles - NR Iterations



Newton-Raphson - Mismatch Convergence



Gauss-Seidel vs Newton-Raphson - 6-Bus System



Known issues of Newton Raphson method

- More sensitive to the starting point.
- Converge issues due to ill-conditioned Jacobian matrix.



Exercise :

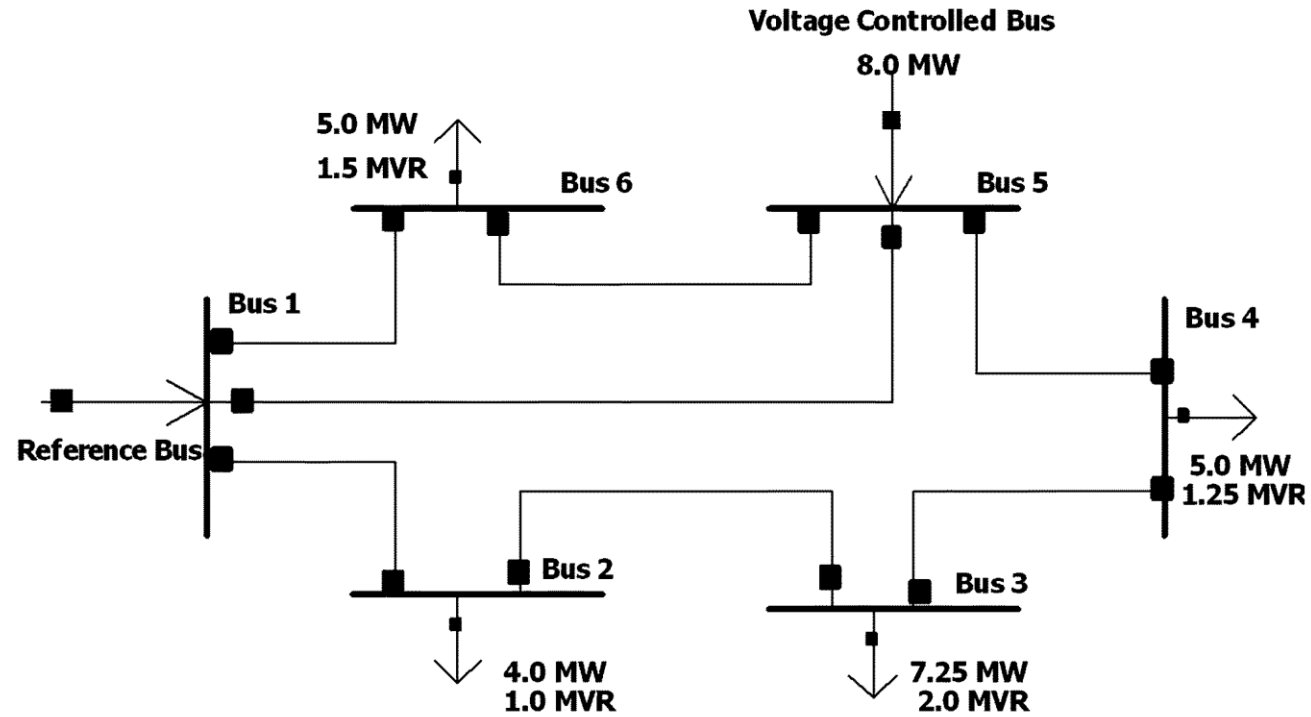


Table 1: Bus Data

Bus No.	Bus Type	Voltage (p.u.)		Bus Power	
		$ V $	δ	(MW)	(MVAR)
1	SB	1	0	Slack bus	
2	PQ	—	—	-4	-1
3	PQ	—	—	-7.25	-2
4	PQ	—	—	-5	-1.25
5	PV	1	—	8	—
6	PQ	—	—	-5	-1.5

Table 2: Line Data

From Bus	To Bus	R_{line} (p.u.)	X_{line} (p.u.)	Y_{shunt} (p.u.)
1	2	0.2238	0.5090	0.0012
2	3	0.2238	0.5090	0.0012
3	4	0.2238	0.5090	0.0012
4	5	0.2238	0.5090	0.0012
5	6	0.2238	0.5090	0.0012
6	1	0.2276	0.2961	0.0025
1	5	0.2603	0.7382	0.0008



Exercise:

The figure shows a 25-kV IEEE 6-bus system which can represent a subtransmission / distribution system. The parameters of this system are given in Table I & II.

Task 1:

- Built the network model using **Pandapower** (Assume a 100MVA base).
- Run a power flow.
- Draw the voltage profile.

Task 2:

- Considering the system, if a 5 MW Photovoltaic plant is planned to be connected at one of the different buses on the system (e.g., at bus 2, 3, 4, 5 or 6), what would be the total power losses in the system for each case?
- What would be the best DG location for minimum system real power losses? Assume the DG operates at unity power factor.



Appendix

Calculation of the Jacobian of a power system

- Taylor Series Approximation
 - Function with one variable
 - Function with multiple variables
 - Multidimensional – multivariable function
- Hadamard product and derivatives rules
- Calculation of the Jacobian

Taylor Series

Example: $f(x) = \sin(x)$

■ A function with one variable

$$f(x) = y$$

Around a point “ a ”:

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f'''(a)}{3!}(x - a)^3 + \dots$$

First order approx. ($x^{(0)} = a$):

$$\Delta f(x) = \left(\frac{df}{dx} \right)^{(0)} \Delta x$$



Multivariable function

$$f(x_1, x_2) = y$$

Example:

$$f(x_1, x_2) = \sin(x_1) \cos(x_2) + 2$$

Taylor approx. around a point “(c,d)”:

$$\begin{aligned} f(x_1, x_2) = & f(c, d) + \left(\frac{\partial f}{\partial x_1}(c, d) \cdot (x_1 - c) + \frac{\partial f}{\partial x_2}(c, d) \cdot (x_2 - d) \right) \\ & + \frac{1}{2!} \left(\frac{\partial^2 f}{\partial x_1^2}(c, d) \cdot (x_1 - c)^2 + 2 \frac{\partial^2 f}{\partial x_1 \partial x_2}(c, d) \cdot (x_1 - c)(x_2 - d) + \frac{\partial^2 f}{\partial x_2^2}(c, d) \cdot (x_2 - d)^2 \right) + \dots \end{aligned}$$

First order approx. ($[x_1^{(0)}, x_2^{(0)}] = [c, d]$):

$$\Delta f(x_1, x_2) = \left[\frac{\partial f}{\partial x_1} \quad \frac{\partial f}{\partial x_2} \right]^{(0)} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \end{bmatrix}$$



Multidimensional multivariable function: First order approximation

$$f_1(x_1, x_2, \dots, x_n) = y_1$$

$$f_2(x_1, x_2, \dots, x_n) = y_2$$

...

$$f_n(x_1, x_2, \dots, x_n) = y_n$$

For small $\Delta x_1, \dots, \Delta x_n$, at $x_1^{(0)}, \dots, x_n^{(0)}$

$$\begin{bmatrix} \Delta f_1(x_1, x_2, \dots, x_n) \\ \Delta f_2(x_1, x_2, \dots, x_n) \\ \vdots \\ \Delta f_n(x_1, x_2, \dots, x_n) \end{bmatrix} = \begin{bmatrix} \left(\frac{df_1}{dx_1}\right)^{(0)} & \left(\frac{df_1}{dx_2}\right)^{(0)} & \dots & \left(\frac{df_1}{dx_n}\right)^{(0)} \\ \left(\frac{df_2}{dx_1}\right)^{(0)} & \left(\frac{df_2}{dx_2}\right)^{(0)} & \dots & \left(\frac{df_2}{dx_n}\right)^{(0)} \\ \vdots & \vdots & \ddots & \vdots \\ \left(\frac{df_n}{dx_1}\right)^{(0)} & \left(\frac{df_n}{dx_2}\right)^{(0)} & \dots & \left(\frac{df_n}{dx_n}\right)^{(0)} \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \vdots \\ \Delta x_n \end{bmatrix}$$



Carl Gustav Jacob Jacobi

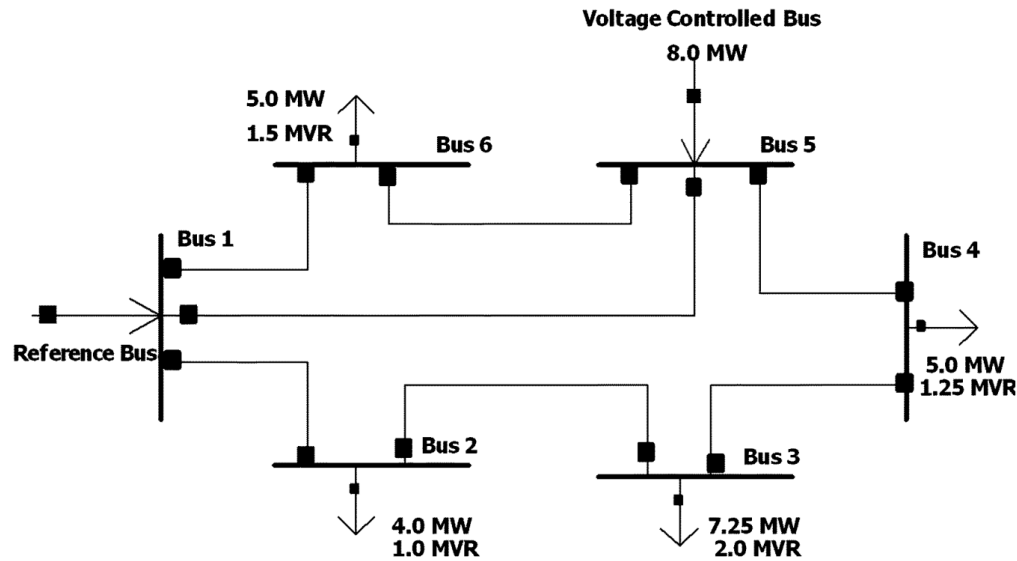
https://en.wikipedia.org/wiki/Carl_Gustav_Jacob_Jacobi

$$\Delta F = \underbrace{J^{(0)}}_{\text{Jacobian at } X^{(0)}} \Delta X$$

Jacobian at $X^{(0)}$



Jacobian in power systems



$$p_1 \rightarrow (v_1, \dots, v_n, \theta_1, \dots, \theta_n)$$

$$q_1 \rightarrow (v_1, \dots, v_n, \theta_1, \dots, \theta_n)$$

⋮

$$q_6 \rightarrow (v_1, \dots, v_n, \theta_1, \dots, \theta_n)$$

$$q_6 \rightarrow (v_1, \dots, v_n, \theta_1, \dots, \theta_n)$$

$$p_i = \sum_{k=1}^n v_i v_k (G_{ik} \cos(\theta_i - \theta_k) + B_{ik} \sin(\theta_i - \theta_k)),$$

$$q_i = \sum_{k=1}^n v_i v_k (G_{ik} \sin(\theta_i - \theta_k) - B_{ik} \cos(\theta_i - \theta_k)).$$

$$\Delta F = J^{(0)} \Delta X$$

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} J \end{bmatrix} \begin{bmatrix} \Delta \theta \\ \Delta V \end{bmatrix}$$

How does this look like for the 6-bus network?



Jacobian calculation: conventional way

$$\begin{bmatrix} \Delta f_1(x_1, x_2, \dots, x_n) \\ \Delta f_2(x_1, x_2, \dots, x_n) \\ \vdots \\ \Delta f_n(x_1, x_2, \dots, x_n) \end{bmatrix} = \begin{bmatrix} \left(\frac{df_1}{dx_1}\right)^{(0)} & \left(\frac{df_1}{dx_2}\right)^{(0)} & \dots & \left(\frac{df_1}{dx_n}\right)^{(0)} \\ \left(\frac{df_2}{dx_1}\right)^{(0)} & \left(\frac{df_2}{dx_2}\right)^{(0)} & \dots & \left(\frac{df_2}{dx_n}\right)^{(0)} \\ \vdots & \vdots & \ddots & \vdots \\ \left(\frac{df_n}{dx_1}\right)^{(0)} & \left(\frac{df_n}{dx_2}\right)^{(0)} & \dots & \left(\frac{df_n}{dx_n}\right)^{(0)} \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \vdots \\ \Delta x_n \end{bmatrix}$$

$$p_i = \sum_{k=1}^n v_i v_k (G_{ik} \cos(\theta_i - \theta_k) + B_{ik} \sin(\theta_i - \theta_k)),$$
$$q_i = \sum_{k=1}^n v_i v_k (G_{ik} \sin(\theta_i - \theta_k) - B_{ik} \cos(\theta_i - \theta_k)).$$

Tedious, inelegant, impractical (not a
vectorize fashion)



Jacobian calculation

$$\bar{s}_i = p_i + j q_i = \bar{v}_i \bar{i}_i^*$$

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} J \end{bmatrix} \begin{bmatrix} \Delta \theta \\ \Delta V \end{bmatrix}$$

- Jacobian relates voltages and power
- Voltage has two variables (magnitude and angle)

$$\bar{v}_i = v_i e^{j\theta_i}$$



Hadamard rules (1/2)

$$\mathbf{C} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

■ Hadamard product

$$\mathbf{F} = [f_{mn}]_{M \times N}, \quad \mathbf{G} = [g_{mn}]_{M \times N}, \quad \mathbf{F} \circ \mathbf{G} = [f_{mn}g_{mn}]_{M \times N}$$

$$\mathbf{C} \circ \mathbf{D} = \begin{bmatrix} 1 \cdot 5 & 2 \cdot 6 \\ 3 \cdot 7 & 4 \cdot 8 \end{bmatrix} = \begin{bmatrix} 5 & 12 \\ 21 & 32 \end{bmatrix}$$

■ Rule 1: $\mathbf{F}, \mathbf{G} \rightarrow \mathbf{X}$

$$D_X(\mathbf{F} \circ \mathbf{G}) = \text{diag}(\mathbf{G})D_X\mathbf{F} + \text{diag}(\mathbf{F})D_X\mathbf{G}.$$

■ Rule 2: $\mathbf{F} \rightarrow \mathbf{X}, \mathbf{A} \rightarrow \text{constant}$

$$D_X(\mathbf{A}\mathbf{F}) = \mathbf{A}D_X(\mathbf{F}).$$



Hadamard rules (2/2)

■ Rule 3: $V \rightarrow$ magnitude, angle

$$D_\delta \mathbf{V} = j \operatorname{diag}(\mathbf{V})$$

$$D_{|V|} \mathbf{V} = \operatorname{diag}(\mathbf{V}) \operatorname{diag}(|\mathbf{V}|^{-1})$$

$$D_\delta \mathbf{V}^* = [j \operatorname{diag}(\mathbf{V})]^*$$

■ Rule 4:

$$\mathbf{F} \circ \mathbf{G} = [f_m g_m]_{M \times 1} = [g_m f_m]_{M \times 1} = \mathbf{G} \circ \mathbf{F}$$

$$\operatorname{diag}(\mathbf{F}) \operatorname{diag}(\mathbf{G}) = \operatorname{diag}([f_m g_m]_{M \times 1}) = \operatorname{diag}([g_m f_m]_{M \times 1}) = \operatorname{diag}(\mathbf{G}) \operatorname{diag}(\mathbf{F})$$



Hadamard rules proof

Introducing a Concise Formulation of the Jacobian Matrix for Newton-Raphson Power Flow Solution in the Engineering Curriculum

Publisher: IEEE

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[Elijah Conlin](#) ; [Niraj Dahal](#) ; [Steven M. Rovnyak](#) ; [James L. Rovnyak](#) [All Authors](#)

<https://ieeexplore.ieee.org/document/9435220/>



Derivative of $\mathbf{S}$ w/r to the angle

$$\mathbf{S} = \mathbf{V} \circ \mathbf{I}^* \qquad \mathbf{I} = \mathbf{Y}\mathbf{V}$$

- With rule 1:

$$D_\delta \mathbf{S} = \text{diag}(\mathbf{I}^*) D_\delta \mathbf{V} + \text{diag}(\mathbf{V}) D_\delta (\mathbf{Y}^* \mathbf{V}^*)$$

- With rule 2-3:

$$D_\delta \mathbf{S} = \text{diag}(\mathbf{I}^*) [j \text{diag}(\mathbf{V})] + \text{diag}(\mathbf{V}) \mathbf{Y}^* [j \text{diag}(\mathbf{V})]^*$$

- With rule 4:

$$D_\delta \mathbf{S} = j \text{diag}(\mathbf{V}) [\text{diag}(\mathbf{Y}^* \mathbf{V}^*) - \mathbf{Y}^* \text{diag}(\mathbf{V}^*)]$$



Derivative of $\mathbf{S}$ w/r to the magnitude

$$\mathbf{S} = \mathbf{V} \circ \mathbf{I}^* \qquad \mathbf{I} = \mathbf{Y}\mathbf{V}$$

- With rule 1:

$$D_{|\mathbf{V}|}\mathbf{S} = \text{diag}(\mathbf{I}^*)D_{|\mathbf{V}|}\mathbf{V} + \text{diag}(\mathbf{V})D_{|\mathbf{V}|}(\mathbf{Y}^*\mathbf{V}^*)$$

- With rule 2-3:

$$D_{|\mathbf{V}|}\mathbf{S} = \text{diag}(\mathbf{I}^*)\text{diag}(\mathbf{V})\text{diag}(|\mathbf{V}|^{-1}) + \text{diag}(\mathbf{V})\mathbf{Y}^*\text{diag}(\mathbf{V}^*)\text{diag}(|\mathbf{V}|^{-1})$$

- With rule 4:

$$D_{|\mathbf{V}|}\mathbf{S} = \text{diag}(\mathbf{V})[\text{diag}(\mathbf{Y}^*\mathbf{V}^*) + \mathbf{Y}^*\text{diag}(\mathbf{V}^*)]\text{diag}(|\mathbf{V}|^{-1})$$



Jacobian submatrices

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} J \end{bmatrix} \begin{bmatrix} \Delta \theta \\ \Delta V \end{bmatrix}$$

$$\mathbf{J} = \begin{pmatrix} \mathbf{J}_{11} & \mathbf{J}_{12} \\ \mathbf{J}_{21} & \mathbf{J}_{22} \end{pmatrix}_{2N \times 2N},$$

$$\mathbf{J}_{11} = \Re(D_{\delta} \Delta \mathbf{S}), \quad \mathbf{J}_{12} = \Re(D_{|\mathbf{V}|} \Delta \mathbf{S}),$$

$$\mathbf{J}_{21} = \Im(D_{\delta} \Delta \mathbf{S}), \quad \mathbf{J}_{22} = \Im(D_{|\mathbf{V}|} \Delta \mathbf{S}).$$

